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Second cycle, 30 credits

Modeling Schizophrenia Symptom Severity: A Deep Learning and Statistical Approach to Substance Use Interactions in Adolescence

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Abstract

Background: Schizophrenia is a multifactorial disorder shaped by genetic, environmental, and behavioral influences, yet how these factors interact during development remains unclear. Adolescent substance use is a key exposure, but its role as an independent risk factor versus an interaction-driven modifier remains debated.

Aim: To investigate how adolescent substance use interacts with genetic liability, adverse childhood experiences, and developmental factors to influence schizophrenia symptom severity, and to assess whether a deep learning framework improves modeling of these interactions.

Methods: Using data from the Swedish Twin Registry (CATSS), we applied Generalized Linear Mixed Models (GLMM), Generalized Additive Mixed Models (GAMM), and a Deep Cross-Modal Fusion Network (DCMFNet). The DCMFNet explicitly models interactions between substance use and multiple modalities. Models were evaluated using error-based and rank-based metrics, with interpretability assessed through feature and interaction importance.

Results: Risk was not driven by single factors but by interacting vulnerabilities. Adverse childhood experiences, prior symptoms, and genetic liability formed a consistent baseline, while substance use modified risk selectively. Positive symptoms showed stronger interaction effects, particularly with ACE and cell-type PRS, whereas negative symptoms were dominated by main effects with weaker interactions. DCMFNet provided modest but consistent improvements, especially in capturing rank-based structure.

Conclusion: Schizophrenia symptom severity is best understood through interacting risk factors. Substance use acts as a context-dependent amplifier rather than an independent driver. DCMFNet offers a flexible framework for modeling such interactions, complementing traditional statistical approaches

Keywords: Schizophrenia, Substance Use, Gene–Environment Interaction, Cell-Type Specific Polygenic Risk Score, Psychotic Symptoms, Machine Learning, Deep Learning, Interaction Modeling

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Contents

List of Abbreviations

PRS	Polygenic Risk Scores
GLMM	Generalized Linear Mixed Model
GAMM	Generalized Additive Mixed Model
DCMFNet	Deep Cross-Modal Fusion Network
GWAS	Genome Wide Association Study
ML	Machine Learning
ACE	Adverse Childhood Experiences
SNP	Single Nucleotide Polymorphism
PCA	Principal Component Analysis
STR	Swedish Twin Registry
CATSS	Child and Adolescent Twin Study in Sweden
LightGBM	Light Gradient Boosting Machine
SHAP	SHapley Additive exPlanations
SUD	Substance Use Disorder
SCZ	Schizophrenia
SES	Socio-Economic Status
ADHD	Attention-Deficit/Hyperactivity Disorder
ASD	Autism Spectrum Disorder
CCA	Complete Case Analysis
MAR	Missing At Random
MNAR	Missing Not At Random
MCAR	Missing Completely At Random
IGF	Iterative Gated Fusion
SE	Squeeze-and-Excitation
ReLU	Rectified Linear Unit
ASGate	Adaptive Selection Gate
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error

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Chapter 1

Introduction

1.1 Background and Motivation

Schizophrenia is a severe psychiatric disorder affecting approximately 1 in 300 individuals worldwide. It is characterized by a combination of positive, negative, and cognitive symptoms. Positive symptoms reflect addition to normal experiences such as hallucinations, delusions, and disorganized thought processes, while negative symptoms reflect a reduction in normal functioning, such as diminished emotional expression and social withdrawal. Cognitive symptoms involve impairments in attention, memory, and executive function [1]. The onset of schizophrenia typically occurs in late adolescence or early adulthood [2]. It is estimated to be 60–80% heritable, making it one of the most strongly genetic mental health disorders [3].

These symptoms arise from disruptions in the underlying neuro-biological systems, particularly involving dopamine signaling across brain regions such as the prefrontal cortex, basal ganglia, and hippocampus [4]. More recent work has shown that these genetic signals are not uniformly distributed across the brain, but are enriched in specific cell types defined by their gene expression profiles [5]. This highlights the importance of considering biological specificity when modeling genetic risk.

Despite this substantial genetic component, genetic predisposition alone does not fully explain the development of the disorder. Twin and family studies demonstrate that many individuals with elevated genetic risk never develop schizophrenia, indicating that environmental and behavioral exposures play a critical role in disease manifestation [3].

Among these exposures, substance use—particularly cannabis—has emerged as one of the most associated risk factors for psychosis. Longitudinal evidence suggests that adolescent cannabis use is linked to an increased risk of psychotic symptoms, especially among individuals with elevated genetic susceptibility [6]. Rather than acting independently, substance use appears to interact with underlying biological vulnerability, reflecting a broader gene–environment interaction framework.

Substance use is therefore treated as the primary exposure in this study. Unlike genetic risk, it is modifiable and varies in timing, frequency, and intensity, particularly during adolescence—a critical period of neuro-development. Since schizophrenia commonly emerges during late adolescence or early adulthood, exposures during this period may have disproportionate effects on disease risk [6].

Moreover, substance use lies at the intersection of multiple domains. It is shaped by socio-economic conditions, adverse childhood experiences, and behavioral traits, while also influencing neuro-biological processes associated with psychotic symptoms. This makes it a natural focal point for studying how genetic, environmental, and behavioral factors jointly contribute to schizophrenia risk [7].

In addition to substance use, several developmental and environmental factors have been associated with schizophrenia. Adverse childhood experiences (ACE), including emotional, physical, and sexual abuse, have been linked to increased risk of later psychopathology [8]. Longitudinal studies further indicate that peer bullying and cannabis use jointly contribute to increasing psychotic experiences over time [9].

Neuro-developmental conditions such as autism spectrum disorder and attention-deficit/hyperactivity disorder are also associated with elevated risk. Individuals with autism are three to six times more likely to develop schizophrenia [10], while individuals with ADHD have a relative risk of 4.3 [11]. Socio-economic status appears to act both as a cause and consequence of schizophrenia [12]. Additionally, sex differences have been observed, with men exhibiting more severe negative symptoms and women showing higher levels of depressive symptoms and functioning [13].

1.2 Research Gap

Although substantial research has examined genetic, environmental, and behavioral risk factors for schizophrenia, fewer studies have investigated how these factors jointly interact during adolescence within a unified analytical framework.

In particular, the combined influence of adolescent substance use, adverse childhood experiences, socio-economic status, and genetic predisposition on psychotic symptoms remains incompletely understood. These relationships are inherently complex and are likely to involve non-linear and higher-order interactions across multiple domains.

This complexity is further amplified in population-based questionnaire data, where variables are high-dimensional, partially correlated, and measured across different developmental stages. In such settings, the number of plausible interactions grows rapidly, and the true structure of these relationships is typically unknown a priori.

As a result, traditional modeling approaches face important limitations. Regression-based models require assumptions about the functional form of relationships and the structure of interactions, which may be difficult to justify in the presence of complex, multi-domain data. This creates a tension between interpretability and flexibility: simpler models may fail to capture relevant patterns, while more flexible models are harder to constrain and interpret.

Together, these challenges highlight the need for modeling approaches that can both accommodate the structure of the data and capture complex interactions without relying on strong prior assumptions.

1.3 Research Aim and Questions

Scientific aim — How does adolescent substance use interact with genetic, environmental, neuro-developmental, and demographic factors to influence schizophrenia symptom severity in early adulthood?

Methodological aim — Can a Deep Cross-Modal Fusion Network better model complex interactions between substance use and all modeled risk factors compared to GLMM and GAMM?

Chapter 2

Theory

2.1 Genetic Risk and Cell-type Polygenic Risk Scores

Polygenic risk scores (PRS) provide a way to summarize an individual's genetic susceptibility to a disorder by aggregating the effects of many genetic variants across the genome. In their standard form, PRS are constructed as weighted sums of risk alleles derived from GWAS summary statistics. Although effective as predictors, conventional PRS are often limited in their interpretability, as they combine signals across the entire genome without distinguishing between underlying biological mechanisms.

To address this limitation, cell-type-specific PRS have been developed to incorporate biological context into genetic risk modeling. This approach leverages gene expression data to identify sets of genes that are specifically active in particular types of brain cells. By restricting PRS calculations to variants associated with these gene sets, it becomes possible to construct risk scores that reflect the specific contributions of cell-types to disease [5].

The gene sets used in this study are derived from the human brain cell atlas [14], which characterizes neuronal populations based on transcriptomic profiles. They represent the top 10% most specifically expressed genes for each cell type enriched for schizophrenia-associated genetic variants. These include excitatory and inhibitory neurons in cortical regions, hippocampal sub-fields involved in memory, amygdala neurons associated with emotional processing, and subcortical populations involved in reward and motor function.

Incorporating cell-type PRS allows the analysis to move beyond a purely predictive framework towards a more mechanistic perspective. It enables the investigation of whether specific biological systems are more strongly associated with psychotic symptoms, and how these genetic signals interact with environmental exposures such as substance use.

2.2 Statistical Modeling Approaches

A natural starting point for analyzing such data is the class of regression models, which aim to quantify relationships between predictors and an outcome variable.

Generalized Linear Models (GLM) extend linear regression by allowing for different types of outcome distributions. However, standard GLMs assume that all observations are independent, which is not appropriate in settings such as twin studies, where individuals are naturally clustered.

Generalized Linear Mixed Models (GLMM) address this limitation by incorporating random effects, which account for correlations within groups, such as twin pairs. In essence, GLMM allows each group to have its own baseline level while estimating overall effects across the population. These models are well-suited for hierarchical data but typically rely on assumptions about linear relationships between predictors and outcomes.

To introduce greater flexibility, Generalized Additive Mixed Models (GAMM) extend GLMM by allowing certain effects to be modeled as smooth, non-linear functions rather than fixed linear terms. This enables the model to capture more complex patterns in the data while still retaining a degree of interpretability. However, in both GLMM and GAMM, interactions between variables must be specified explicitly, and the form of these interactions is determined by the researcher.

2.3 Machine Learning and Deep Learning Approaches

As the number of predictors and potential interactions increases, explicitly specifying all relevant relationships becomes increasingly challenging. Machine learning approaches provide an alternative by learning patterns directly from the data, rather than relying on pre-defined functional forms.

In this context, machine learning refers to a class of algorithms that optimize predictive performance by identifying patterns in the data through training. These models can capture non-linear relationships and interactions implicitly, making them particularly useful in high-dimensional settings. However, they often operate as “black-box” models, meaning that the learned relationships are not easily interpretable.

Deep learning represents a subset of machine learning based on neural networks with multiple layers. These models learn hierarchical representations of data, where simple patterns are combined to form increasingly complex features. This layered structure enables deep learning models to capture high-order interactions and complex dependencies between variables.

Machine learning and deep learning approaches are increasingly explored in mental health research for integrating high-dimensional and multi-domain data. For instance, prior work using CATSS data applied several machine learning models to predict adolescent mental health outcomes, demonstrating their potential while showing comparable performance to traditional regression approaches [15].

This suggests that while these methods can capture complex relationships, their practical advantage depends on the underlying structure of the data.

Among deep learning architectures, models designed for cross fusion of multimodal data are particularly relevant in this study. The Deep Cross-Modal Fusion Network (DCMFNet) was originally proposed to integrate heterogeneous data sources, such as images and text, by iteratively combining information across modalities [16]. The core idea is to allow different groups of features to interact in a structured manner, rather than treating all variables as a single homogeneous set.

In this study, all the genetic, environmental, and behavioral variables are treated as distinct modalities within the DCMFNet framework. This allows the model to learn how these domains interact with one another, capturing complex and potentially non-linear relationships that are difficult to specify explicitly in traditional models.

To evaluate the effectiveness of this approach, it is compared with traditional statistical models, including Generalized Linear Mixed Models (GLMM) and Generalized Additive Mixed Models (GAMM).

The primary focus of this study is on comparing interpretable statistical models with flexible deep learning approaches in modeling complex interactions. To enable such a comparison, it is necessary to first establish a coherent and well-structured dataset that reflects the interplay of genetic, environmental, and behavioral factors over time.

Chapter 3

Data

3.1 Study Cohort

This study uses data from the Swedish Twin Registry (STR), specifically the Child and Adolescent Twin Study in Sweden (CATSS), which provides cross-sectional questionnaire-based data at pre-defined ages 9, 15, and 18. This design enables the analysis of developmental relationships between early-life exposures and later mental health outcomes.

The primary exposure is substance use at age 15, and the outcomes are psychotic and affective symptoms at age 18. Positive symptoms include psychotic and manic features, while negative symptoms are proxied by depressive symptoms.

3.2 Genotype Data and Cell-type PRS Generation

Genotype data for CATSS individuals were obtained from the STR, and cell-type PRS were computed using the PRS-CS framework [17]. The selection of individuals for genotype analysis is summarized in Figure 3.1. This process ensures that only individuals with complete and reliable genotype information are included in downstream analyses.

3.2.1 Cell-type PRS Pipeline

The PRS pipeline, adapted from Wang et al. [18], involves quality control, SNP annotation, genotype cleaning, and PRS computation across chromosomes. Specifically, SNP effect sizes are estimated using a Bayesian framework (PRS-CS) [1] and aggregated to generate individual-level risk scores for each cell type. This systematic pipeline ensures that the resulting PRS are robust, biologically informed, and comparable across individuals.

Correlation analysis of the resulting PRS revealed clustering patterns across neuronal groups. This is visualized in Figure 3.2, where excitatory, inhibitory, and spiny neuron PRS

[1] https://github.com/getian107/PRScs/tree/master

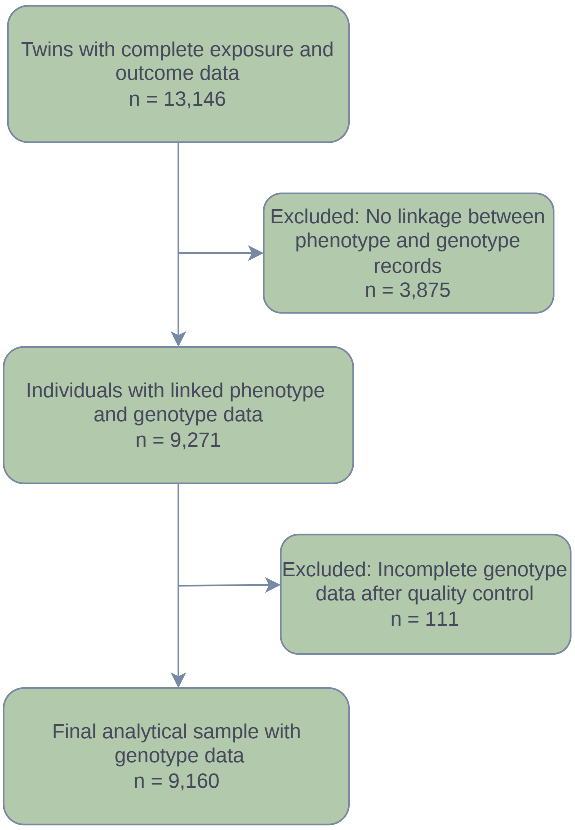


Figure 3.1: Flow chart of genotype data processing for cell-type PRS generation.

show within-group correlation structures, suggesting shared genetic architecture. Overall, moderate positive correlations are observed across most cell-type PRS, reflecting a shared polygenic architecture underlying schizophrenia risk. The relatively stronger correlations within biologically related neuronal groups suggest that these scores capture overlapping functional pathways. Although, no pairwise correlation exceeds 0.7, indicating that strong multicollinearity is not present among the predictors. This supports the inclusion of all cell-type PRS in subsequent analyses, as each score is likely to contribute complementary information without compromising model stability.

Genetic confounding: To account for population structure and batch effects, principal component analysis (PCA) was applied. PRS were adjusted using batch-specific principal components, and individuals with non-European ancestry were identified as potential outliers.

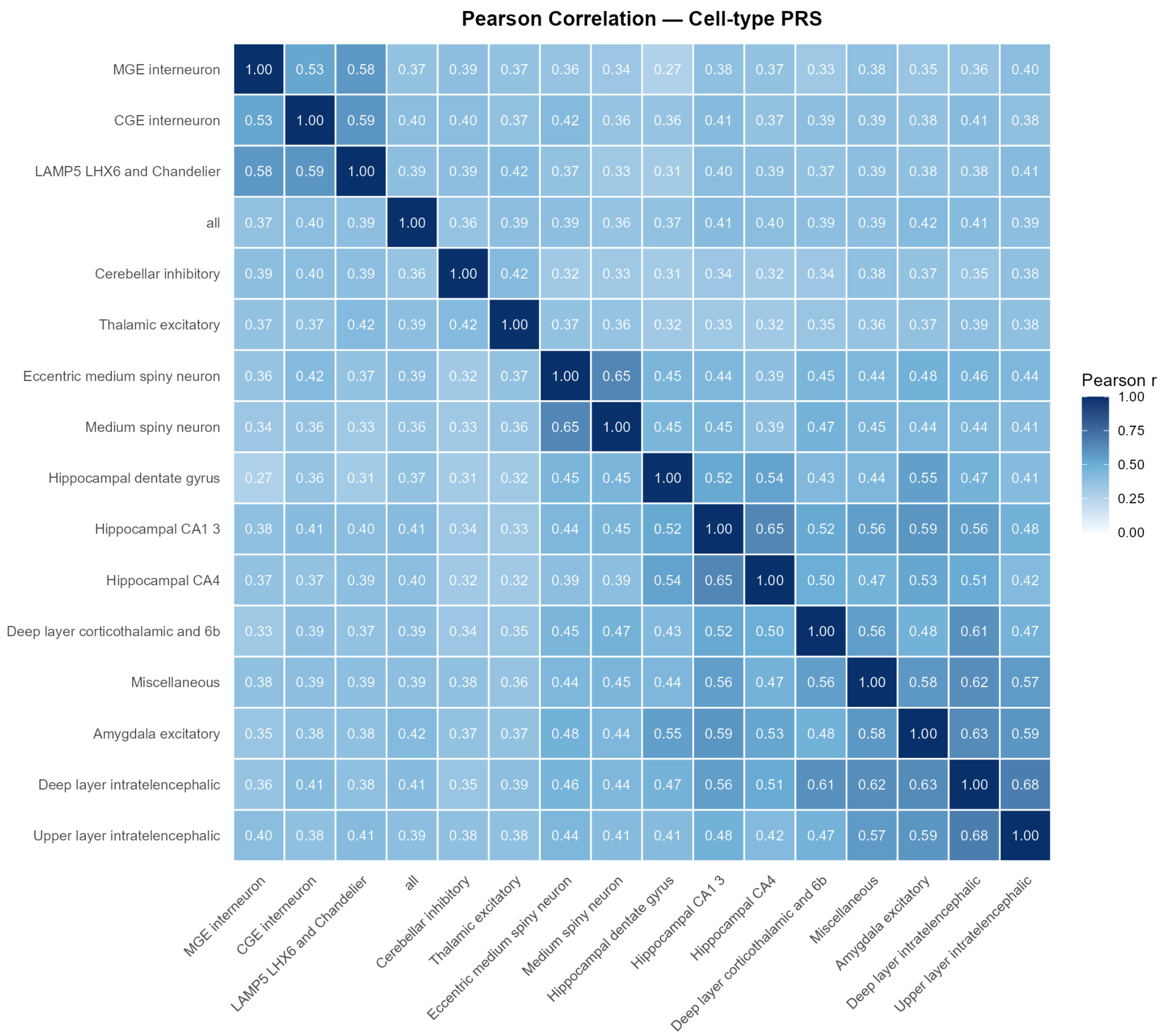


Figure 3.2: Pearson correlation matrix of cell-type PRS

3.3 Phenotype Data

The phenotype data includes behavioral, environmental, and clinical variables derived from questionnaires across multiple time points. The dataset contains over 200 variables, many of which are correlated due to overlapping constructs and measurement scales.

The variables included in this study span multiple domains:

- **Exposure:** Substance use at age 15, including cannabis, other drugs, smoking, alcohol, and snuff.
- **Environmental factors:** Adverse Childhood Experiences (ACE) at ages 9, 15, and 18, including abuse, bullying, neglect, and exposure to violence, along with early ADHD and autism-related symptoms at age 9.
- **Demographic factors:** Socio-economic status (SES), parental education, country of birth, and sex.

- **Baseline symptoms:** Psychotic, manic, and depressive symptoms at age 15.
- **Outcomes:** Psychotic, manic, and depressive symptoms at age 18.

These variables are derived from structured clinical questionnaires and are inherently high-dimensional, with each construct represented by multiple symptom-level indicators.

Positive and negative symptoms are analyzed separately due to their fundamentally different underlying structures and risk mechanisms.

Substance use and schizophrenia-related symptoms are both measured at age 15; due to unclear temporal ordering, baseline symptoms (SCZ15) are included as an adjustment variable, acknowledging their potential role as confounders or mediators.

Initial descriptive analyses were conducted on all variables, but high dimensionality and multicollinearity limited interpretability; feature selection was therefore applied to reduce complexity.

3.4 Feature Selection

Feature selection was conducted as a structured pipeline combining statistical exploration and model-based approaches. First, pairwise Pearson correlations were examined to identify redundancy and inform grouping of related variables. Second, a LightGBM model was used to evaluate feature importance. LightGBM is a gradient boosting framework that combines multiple decision trees to model complex relationships efficiently [19]. Its tree-based structure naturally captures non-linear effects and interactions, making it well-suited for exploratory feature screening in high-dimensional data.

Feature importance was interpreted using SHAP (TreeSHAP) values, which quantify the contribution of each feature to model predictions. SHAP clustering was used to identify groups of variables with similar influence patterns (see Appendix A).

The overall feature selection workflow is summarised in Figure 3.3, which outlines the sequential filtering and grouping steps.

The following criteria were applied for feature selection:

- Removal of variables with very high missingness ($> 90\%$)
- Grouping or removal of highly correlated variables (> 0.9)
- Use of SHAP-based clustering for moderately correlated features
- Removal of variables with 50–90% missingness when not informative

Based on these criteria, highly correlated substance use variables were aggregated, parent-reported outcomes were removed due to higher missingness, and ACE-related variables were grouped to reduce redundancy.

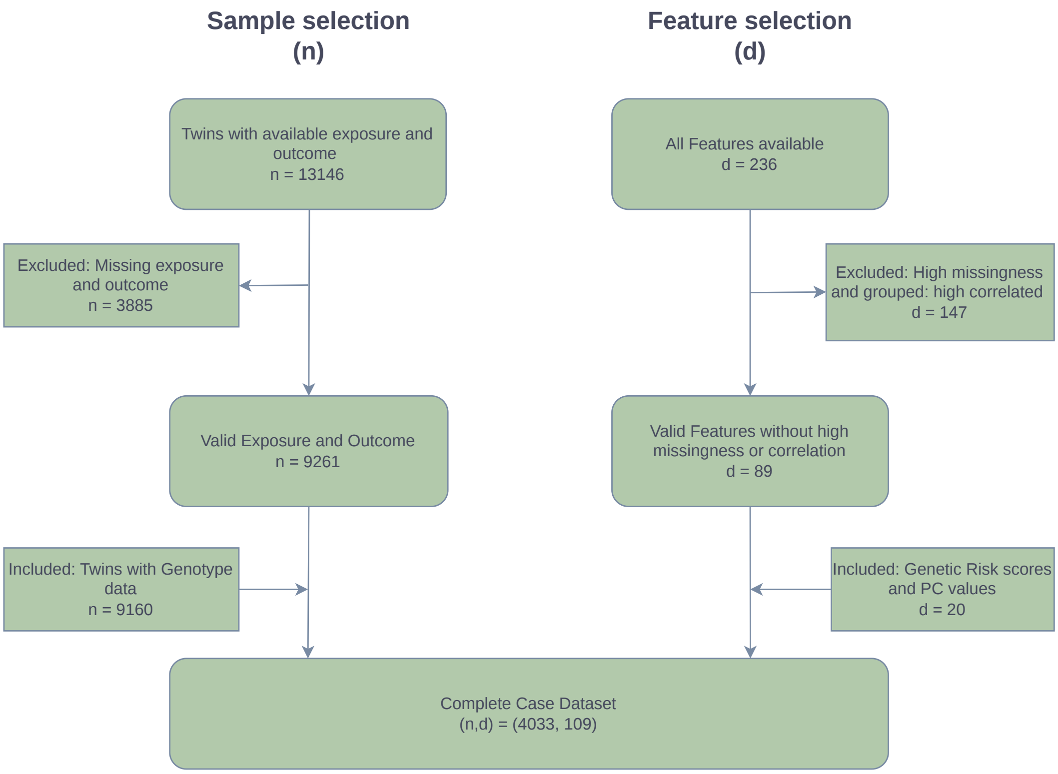
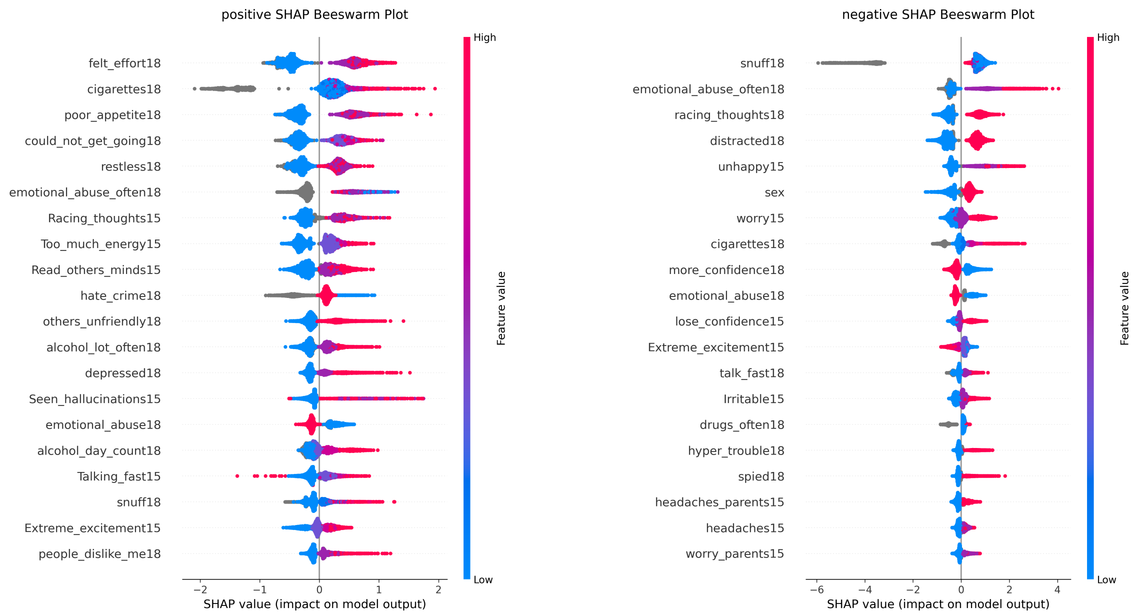


Figure 3.3: Feature selection workflow

For instance, the `OtherDrugs15` variable aggregates multiple highly correlated substances, including amphetamines, cocaine, opioids (e.g., heroin, morphine), psychedelics (e.g., LSD, mushrooms), inhalants, and performance-enhancing drugs. Outcome variables were constructed as normalized aggregated symptom scores between 0 and 1, preserving variability while avoiding arbitrary thresholds.

Feature importance patterns are visualized in Figure 3.4 using SHAP beeswarm plots, where each point represents the contribution of a feature to an individual prediction. The horizontal position indicates whether the feature increases (right) or decreases (left) the predicted outcome, while color reflects the feature value (red = high, blue = low). The spread of points along the x-axis reflects the variability and overall influence of each feature.

Features such as substance use (e.g., cigarettes and snuff at age 18), emotional abuse, and affective symptoms (e.g., low mood, restlessness) show consistent positive contributions to predicted risk. Additionally, baseline psychological features at age 15 (e.g., racing thoughts, worry, confidence) contribute meaningfully, highlighting the role of early mental health signals in later outcomes.



(a) Positive model (b) Negative model

Figure 3.4: SHAP-based feature importance for positive and negative outcomes

3.5 Descriptive Statistics

Table 3.1 summarizes the key modalities included in the analysis, highlighting a mix of ordinal, binary, and continuous variables spanning behavioral, environmental, and genetic domains.

Overall, most variables exhibit low to moderate missingness, although earlier-life measures such as SCZ15 and ADHD9 show higher proportions of missing data, reflecting challenges in data collection over the years.

The outcome variables (SCZ18 positive and negative symptoms) are continuous and skewed towards lower values, consistent with the low prevalence of high-risk cases observed in the cohort, as also seen in Figure 3.5 and 3.6.

Modality	N Vars	Type	Mean ± SD	Median	Range	Missing
SUD15 (Substance Use, age 15)	6	Ordinal	0.706 ± 1.388	0.000	0.000–5.000	0.0%
PRS (Polygenic Risk Scores)	16	Continuous	-0.085 ± 0.276	-0.044	-2.648–0.668	28.3%
SCZ15 (Psychotic Experiences, age 15)	23	Ordinal	0.448 ± 0.733	0.000	0.000–3.000	27.4%
ADHD9 (ADHD Symptoms, age 9)	19	Ordinal	0.160 ± 0.443	0.000	0.000–2.000	14.9%
ASD9 (ASD Traits, age 9)	17	Ordinal	0.071 ± 0.303	0.000	0.000–2.000	15.9%
ACE15 (Adverse Experiences, age 15)	7	Ordinal	1.219 ± 0.693	1.000	1.000–6.000	4.6%
ACE18 (Adverse Experiences, age 18)	4	Binary	88.7%	NA	0, 1	11.1%
SUD18 (Substance Use, age 18)	4	Ordinal	0.925 ± 1.479	0.000	0.000–7.000	0.3%
SES (Socioeconomic Status)	4	Ordinal	2.550 ± 1.097	2.000	0.000–5.000	18.0%
SEX	1	Categorical	Male: 3221 (34.8%), Female: 4683 (50.6%)	NA	1, 2	14.7%
SCZ18 Positive (Outcome)	1	Continuous	0.139 ± 0.106	0.130	0.000–0.870	0.0%
SCZ18 Negative (Outcome)	1	Continuous	0.256 ± 0.181	0.212	0.000–1.000	0.0%

Table 3.1: Descriptive statistics of input modalities and outcome variables.

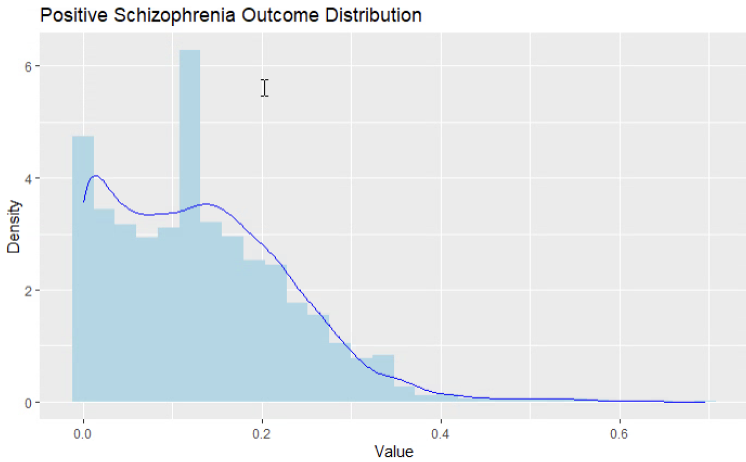


Figure 3.5: Positive symptoms Outcome Distribution

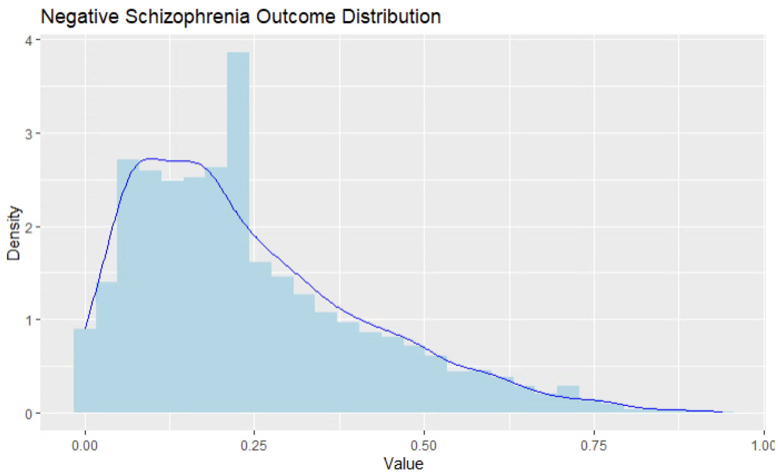


Figure 3.6: Negative symptoms Outcome Distribution

3.6 Handling Missing Data

The missingness mechanism could not be clearly classified as MCAR, MAR, or MNAR. Key variables (e.g., ACE9, SCZ15) are temporally prior to most covariates, limiting the feasibility of reliable imputation without strong assumptions. Parental-reported variables also exhibited substantial missingness and could not be reconstructed.

Complete case analysis was therefore adopted as a pragmatic approach. While this may introduce selection bias if missingness is not random, it avoids imposing strong and unverifiable assumptions required for imputation.

Sparse responses (e.g., `Don't know/Don't want to answer`) were treated as missing to maintain consistency.

Chapter 4

Methodology

4.1 Modeling Strategy

Two separate models were constructed for the two outcomes of interest: positive and negative psychotic symptoms at age 18. Each modeling approach was applied consistently across both outcomes to ensure comparability.

The dataset was split into training and testing sets using a 75–25 split, while preserving the distribution of the outcome variables. To avoid data leakage and account for the twin structure, both individuals from a twin pair were always assigned to the same split.

4.2 Statistical Models

4.2.1 Generalized Linear Mixed Model (GLMM)

We first implemented a GLMM with psychotic symptoms as the outcome variable. All environmental, genetic, and behavioral variables were included as covariates. The outcome was modeled using a Gaussian distribution with an identity link, consistent with the continuous nature of the constructed symptom scores.

To explicitly study interactions with the primary exposure, interaction terms between `SUD15` and all other covariates were included. Genetic confounding was accounted for by including $batch \times PC$ terms. The twin structure was incorporated through a random intercept for each twin pair denoted by `cmpair`.

The model can be expressed as:

$$\begin{aligned} \texttt{SCZ18} \sim \texttt{covariate\_terms} + \texttt{interaction\_terms} + \\ \texttt{genetic\_confounding\_terms} + (1 \mid \texttt{cmpair}) \end{aligned}$$

This formulation provides a structured and interpretable baseline, allowing us to estimate both main effects and pre-specified interactions while accounting for within-pair dependence in the twin data.

4.2.2 Generalized Additive Mixed Model (GAMM)

To relax the assumption of linearity in the GLMM, we implemented a GAMM. The model specification is similar to the GLMM, but allows for non-linear relationships through smooth functions, applied here to each cell-type PRS variable to avoid assuming a strictly linear relationship between genetic risk and psychotic outcomes.

The model is specified as:

$$\begin{aligned}\texttt{SCZ18} \sim \texttt{covariate\_terms} + s(\texttt{PRS}_k) + \texttt{interaction\_terms} + \\ \texttt{genetic\_confounding\_terms} + s(\texttt{cmpair}, bs =' re')\end{aligned}$$

Taken together, GLMM and GAMM provide complementary statistical baselines: the former offers a strictly parametric framework, while the latter introduces limited flexibility through smooth functions and extends to capture moderate non-linear effects.

While, the GAMM relaxes linearity for individual PRS terms, it cannot capture higher-order interactions between covariates or learn latent representations of combined exposures — motivating the use of a deep cross-modal fusion approach.

4.3 Deep Learning Model

4.3.1 DCMFNet Architecture

The Deep Cross-Modal Fusion Network (DCMFNet) is designed to explicitly model interactions between a primary exposure and multiple covariates through iterative cross-feature fusion. In this study, the architecture consists of 9 Iterative Gated Fusion (IGF) blocks, where each block models the interaction between the primary exposure ($SUD15$) and a single covariate. The overall architecture is illustrated in Figure 4.1.

Within each IGF block, the guided feature ($SUD15$) and a guidance feature (covariate) are fused iteratively across multiple layers (set to 5 by default, following [16]). This process progressively refines the interaction representation through repeated transformation and filtering.

The fusion update at each layer is defined as:

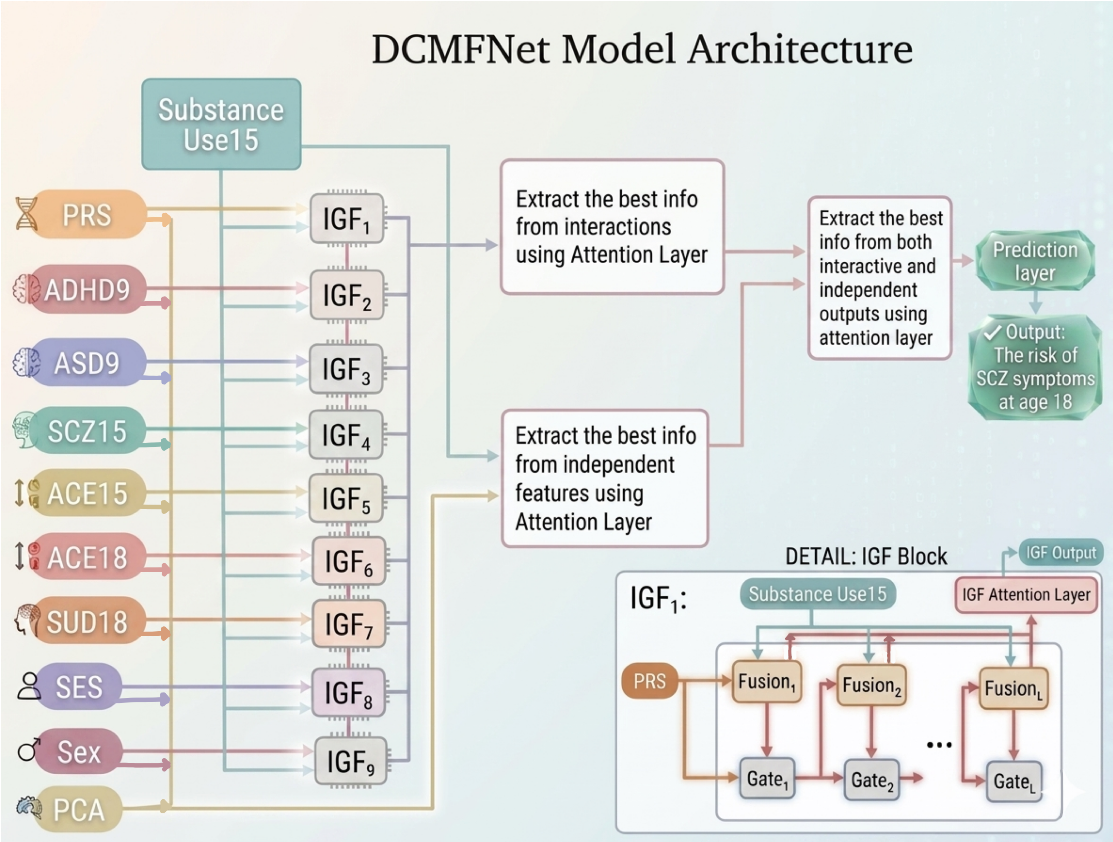


Figure 4.1: Overview of the adapted DCMFNet architecture for tabular genotype and phenotype data.

$$F_{curr} = \tau(\tau(W_1 x) \odot \tau(W_2 G_{prev}))$$

where τ denotes the tanh activation, $\odot$ represents element-wise multiplication, and $W_1 \in \mathbb{R}^{N_x \times N_m}$ and $W_2 \in \mathbb{R}^{N_m \times N_m}$ are learnable weight matrices. Here, x corresponds to the guided feature ($SUD15$), and $G_{prev} \in \mathbb{R}^{b \times N_m}$ is the output of the gating mechanism from the previous layer, with b denoting the batch size.

To mitigate noise introduced during iterative fusion, an adaptive selection gate (ASGate) is applied:

$$\begin{aligned}
z_t &= \sigma(W_z(F_t + G_{t-1}) + b_z) \\
f_t &= F_t \odot z_t \\
G_t &= \tau(W_g([F_t; f_t]) + b_g)
\end{aligned}$$

where $W_z \in \mathbb{R}^{N_m \times N_m}$ and $W_g \in \mathbb{R}^{2N_m \times N_m}$ are learnable parameters, b_z and b_g are bias terms, and σ denotes the sigmoid activation.

This gating mechanism selectively amplifies informative signals while suppressing noise, which is particularly important in high-dimensional settings where spurious interactions may otherwise dominate the learned representations.

4.3.2 Model Adaptation

Attention Mechanism

Tabular data lacks inherent spatial or sequential structure, making convolution-based operations unsuitable. We therefore employ a channel-wise attention mechanism to capture dependencies between features and contextual understanding for the model.

We adopt the Squeeze-and-Excitation (SE) mechanism [20], which recalibrates feature representations by modeling inter-feature relationships. For an intermediate representation $X \in \mathbb{R}^{b \times C}$, where b is the batch size during training and C the number of features, the SE block operates as follows:

Squeeze: A global summary is computed for each feature:

$$z_c = \frac{1}{b} \sum_{i=1}^{b} X_{i,c}$$

Excitation: Feature dependencies are captured via a gating function:

$$s = \sigma(W_2 \cdot \delta(W_1 \cdot z))$$

where W_1 and W_2 are learnable parameters and δ denotes a non-linear activation (ReLU).

Recalibration: The features are rescaled as:

$$\tilde{X}_{i,c} = s_c \cdot X_{i,c}$$

This mechanism emphasizes informative features while suppressing less relevant ones. Unlike soft attention, which assigns weights at the instance level, SE attention captures global feature importance, leading to improved stability and reduced overfitting in this setting.

Loss Function

The outcome variables exhibit a highly skewed distribution, with a small number of high-risk observations. To address this imbalance in a regression setting, we employ a composite loss function that combines robust error estimation with adaptive reweighting.

Base Loss (Huber Loss): The Huber loss is defined as:

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & \text{if } |y - \hat{y}| \leq \delta \\ \delta \left(|y - \hat{y}| - \frac{1}{2}\delta \right), & \text{otherwise} \end{cases}$$

This loss behaves like Mean Squared Error (MSE) for small residuals and Mean Absolute Error (MAE) for large residuals, providing robustness to outliers.

Inverse-Frequency Weighting: Each observation is assigned a weight based on the inverse frequency of its target bin:

$$w_i = \frac{1}{\text{freq}(y_i)}$$

This ensures that rare high-value outcomes contribute more to the loss.

Focal Regression Component: To further emphasize difficult-to-predict observations, a focal weighting term is applied:

$$L_{focal} = w_i \cdot |y_i - \hat{y}_i|^\gamma \cdot L_\delta(y_i, \hat{y}_i)$$

where $\gamma > 1$ controls the degree of emphasis on larger errors. This formulation prioritizes samples with higher residuals, which are often underrepresented tail cases.

The final loss is computed as the weighted sum across all samples.

4.4 Machine Learning Benchmarks

To contextualize the model performances, we include a range of standard machine learning models implemented using the `lazypredict` package [21], which enables rapid evaluation of multiple regression algorithms under a consistent framework. This includes models from different families such as linear, regularized, tree-based, ensemble, kernel, and nearest-neighbor methods.

These models were trained without hyperparameter tuning to provide an out-of-the-box benchmark of predictive performance. These models serve as empirical benchmarks rather than objects of interpretation, as their interactions are learned implicitly and cannot be explicitly specified, making direct comparison with structured models such as GLMM, GAMM, and DCMFNet less meaningful.

4.5 Model Training and Evaluation

All models were trained using the same train–test split to ensure a fair comparison. Model performance was evaluated on the test set using residual R^2, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Pearson correlation (r), and Spearman correlation (ρ).

Residual R^2 quantifies the proportion of variance explained by the model, RMSE and MAE measure the magnitude of prediction error (with RMSE penalizing larger errors more strongly), while Pearson r and Spearman ρ assess linear and rank-based agreement between predicted and observed values, respectively. In particular, correlation-based metrics are included to assess whether the models capture the relative ordering of risk, which is important in the presence of skewed outcome distributions.

Together, these metrics provide complementary perspectives on predictive performance, but do not assess causal relationships, model calibration, or the correctness of inferred interaction structures.

The following chapter presents the results and evaluates the relative performance of the proposed modeling approaches.

Chapter 5

Results and Analysis

5.1 GLMM

We first examine the GLMM results for both positive and negative psychotic symptom outcomes.

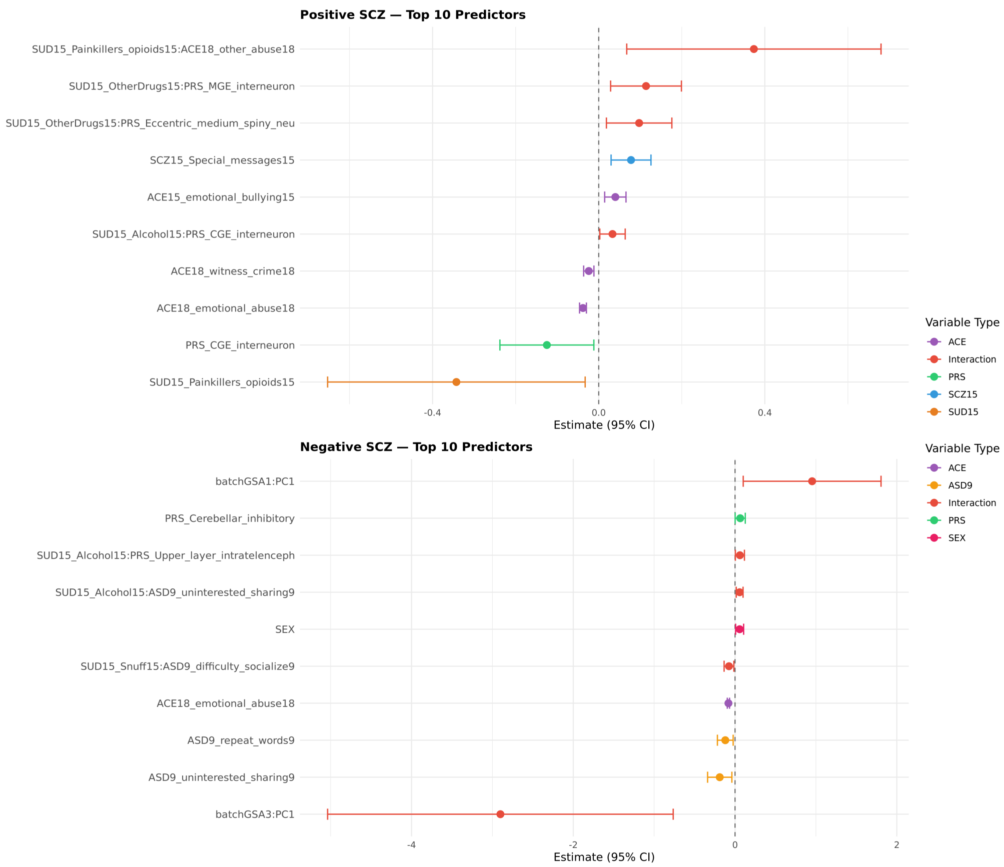


Figure 5.1: Top 10 predictors for positive and negative psychotic symptoms based on GLMM regression coefficients (95% confidence intervals) at seed 42.

Figure 5.1 shows the top predictors ranked by effect size. For positive symptoms, interaction terms between substance use and environmental or genetic factors dominate. In particular, interactions involving substance use with adverse experiences and cell-type PRS show the strongest positive associations. Most confidence intervals do not cross zero, indicating statistically meaningful effects. The interaction between opioid use and ACE18 is positive, while the main effect of opioid use is negative. This suggests that in the absence of adverse childhood experiences at age 18, opioid use is associated with lower symptom severity, but in the presence of ACE18, this relationship reverses, indicating a context-dependent effect.

For negative symptoms, the structure differs, with stronger contributions from genetic components and confounding terms (e.g., batch × PC), and weaker interaction effects.

Model diagnostics support the specification, with QQ plots of twin-level random effects indicating approximate normality (Appendix C.1).

We next examined main and interaction effects separately for positive and negative symptom models.

For positive symptoms, main effects are generally small and largely dominated by baseline symptom measures (SCZ15), as shown in Figure 5.2. This suggests that individual predictors contribute weakly when considered in isolation.

In contrast, the negative symptom model exhibits more diffuse main effects, indicating that the predictive signal is distributed across multiple features rather than driven by a few strong effects (Figure 5.3). Moreover, for negative symptoms, the signal is primarily captured by main effects—particularly PRS, sex, SCZ15, and ASD—while interaction effects play a comparatively minor role.

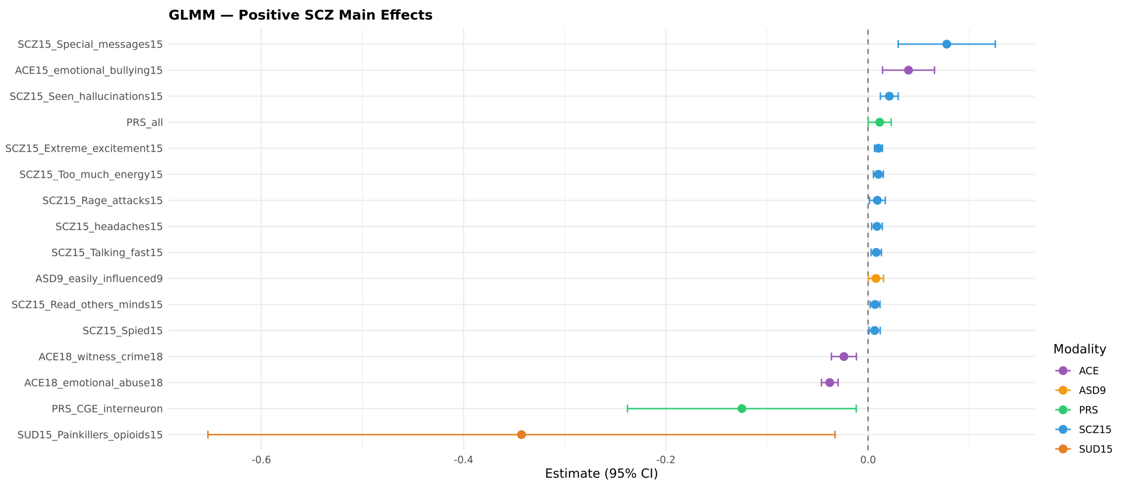


Figure 5.2: Main effects in Positive GLMM model at seed 42 using regression coefficients.

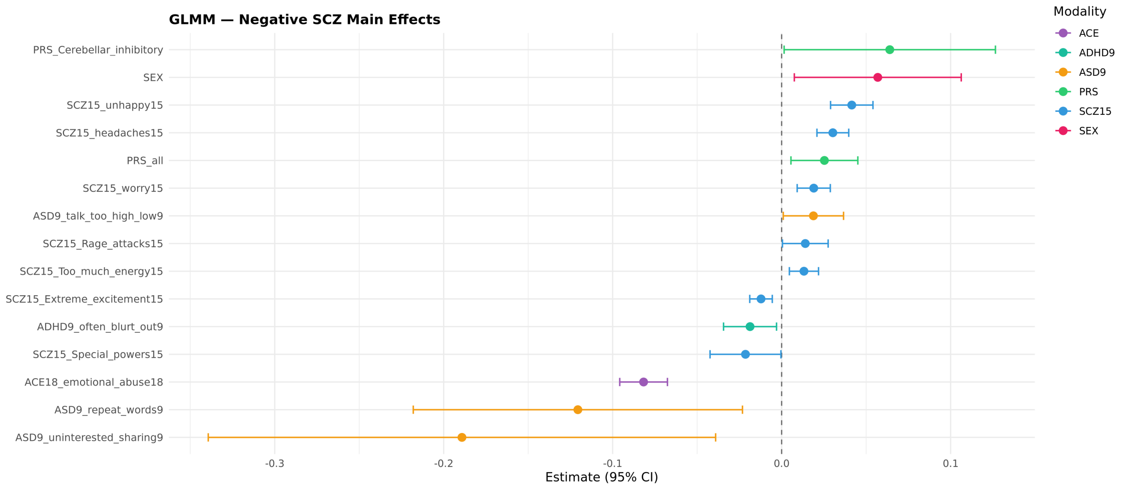


Figure 5.3: Main effects in Negative GLMM model at seed 42 using regression coefficients.

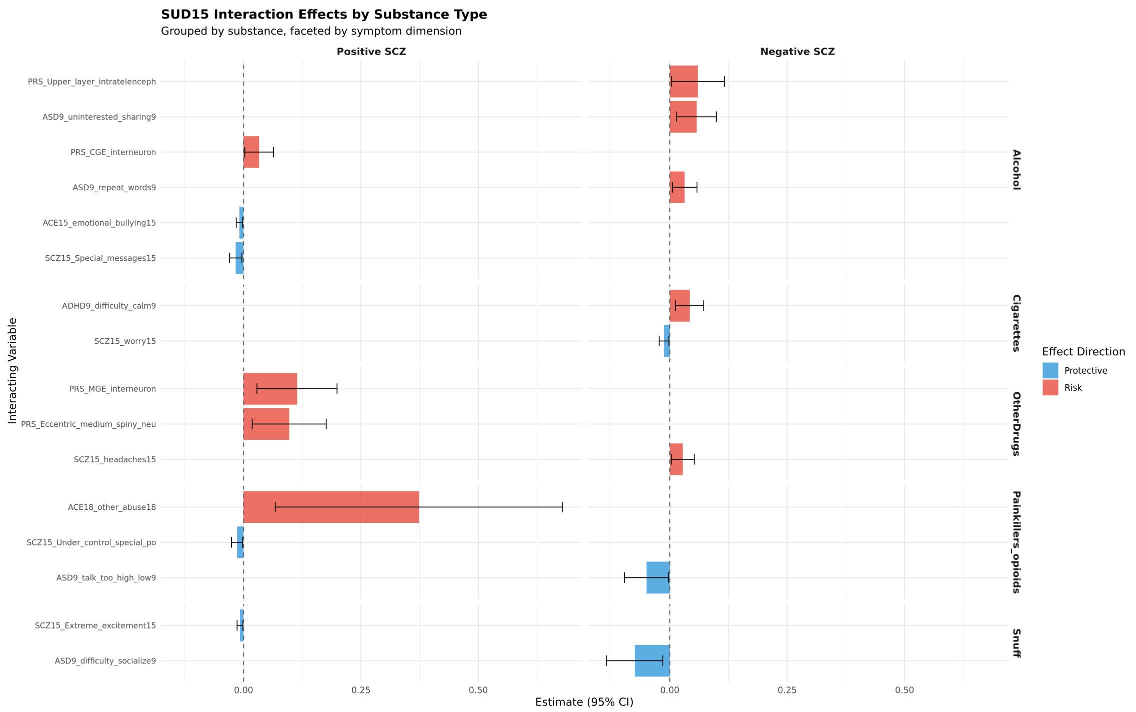


Figure 5.4: Interaction effects between substance use and covariates in GLMM models, stratified by substance type at seed 42 using regression coefficients.

The interaction structure (Figure 5.4) shows heterogeneous patterns. Interactions involving *other drugs* and PRS components are consistently positive, while opioid use shows strong interactions with severe adverse experiences (e.g., $ACE18\_other\_abuse18$). Other substances, such as snuff, exhibit weaker or negative interactions.

For negative symptoms, interaction effects are weaker and more dispersed, indicating a less interaction-driven structure.

Overall, these results suggest that substance use contributes to risk primarily through selective interactions with environmental and genetic factors.

5.2 GAMM

We next examine the GAMM results for both positive and negative psychotic symptom outcomes.

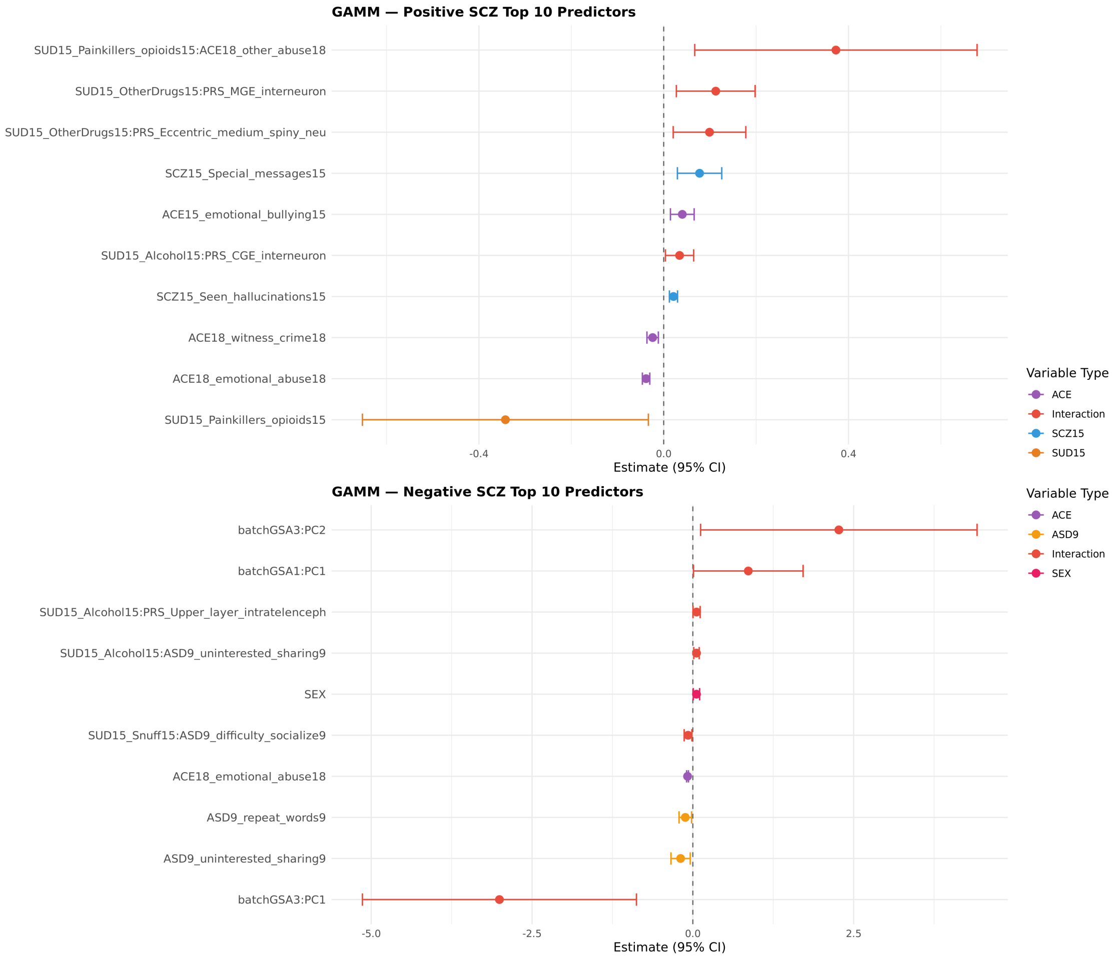


Figure 5.5: Top 10 predictors for positive and negative psychotic symptoms based on GAMM regression coefficients (95% confidence intervals) at seed 42.

Figure 5.5 shows the top predictors ranked by effect size. Consistent with GLMM, interaction terms between substance use and environmental or genetic factors dominate the positive symptom model, particularly opioid–ACE and substance use–PRS interactions.

Allowing for non-linearity introduces only minor changes. PRS effects are modeled as smooth terms and therefore do not appear among main effects; examination of these terms shows that the aggregate PRS (PRS_all) is statistically significant in both models, with some cell-type specific PRS showing marginal effects (e.g., CGE interneuron and hippocampal CA1–3 for positive symptoms, and cerebellar inhibitory and upper-layer intratelencephalic for negative symptoms) as seen in Table 5.1. Most smooth terms remain weak and close to linear.

Outcome	Term	edf	F	p-value	Sig.
Positive	$s(PRS\_all)$	1.00	4.01	0.045	*
	$s(PRS\_CGE\_interneuron)$	2.01	2.91	0.056	.
	$s(PRS\_Hippocampal\_CA1\_3)$	2.60	2.19	0.082	.
	$s(cmpair)$	348.25	0.22	<0.001	***
Negative	$s(PRS\_all)$	1.00	6.50	0.011	*
	$s(PRS\_Cerebellar\_inhibitory)$	2.10	2.18	0.076	.
	$s(PRS\_Upper\_layer\_intratelencephalic)$	1.00	3.50	0.062	.
	$s(cmpair)$	385.15	0.24	<0.001	***

Table 5.1: Statistically significant and marginally significant smooth terms in GAMM models for positive and negative outcomes ($p < 0.1$). edf = estimated degrees of freedom.

Note: Significance levels: *** $p < 0.001$, * $p < 0.05$, . $p < 0.1$.

For negative symptoms, the structure closely mirrors GLMM, with genetic and confounding terms (e.g., batch × PC) contributing more than interaction effects.

Model diagnostics indicate adequate fit, with approximately normal twin-level random effects (Appendix C.1).

Interaction patterns are not shown for GAMM, as they are unchanged from GLMM. Overall, both main (see Appendix C.2) and interaction effects remain largely consistent with GLMM, indicating that non-linear extensions provide limited additional signal.

5.3 DCMFNet

5.3.1 Default Model Training

The DCMFNet model was first trained using the default hyperparameters reported in [16]. An internal 80–20 train–validation split was used during training, and the final model was evaluated on the held-out test set.

Across both positive and negative symptom models as seen in Figure 5.6 and 5.7, training loss decreases rapidly in the initial epochs and then stabilizes, indicating effective optimization. Validation MAE follows a similar trend, with a small but consistent gap relative to training MAE, suggesting mild but controlled overfitting.

Spearman correlation increases steadily during training for both models, reaching approximately 0.46–0.50, indicating that the model is able to capture meaningful rank-based

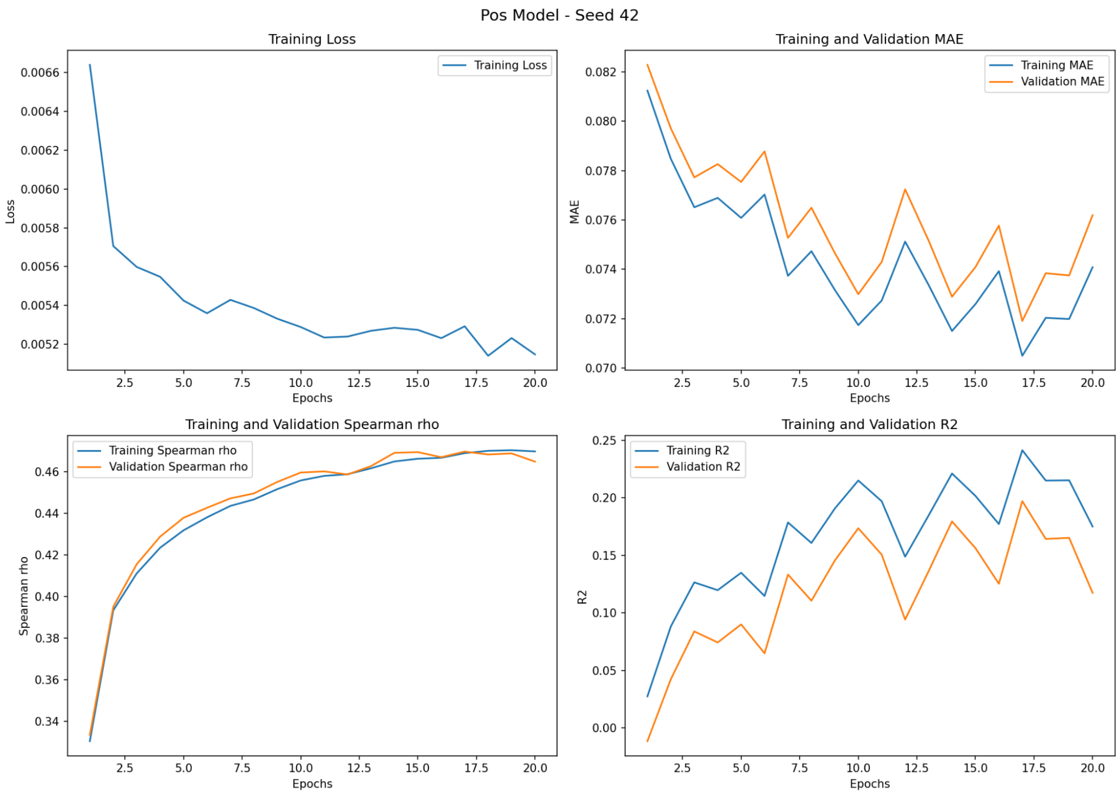


Figure 5.6: Training dynamics for the positive symptom model under default hyperparameters (seed 42).

structure in the outcomes. Similarly, R^2 improves over epochs but remains modest (approximately 0.15–0.20 on validation), reflecting limited explained variance despite stable learning.

Notably, validation curves exhibit fluctuations across epochs, particularly for R^2 and MAE, indicating sensitivity to the default hyperparameter configuration. This suggests that while the model converges, its generalization performance is not fully optimized, motivating further hyperparameter tuning.

5.3.2 Hyperparameter Tuning

To improve model generalization, hyperparameters were optimized using Optuna, including the number of IGF layers. Optuna performs efficient hyperparameter search using Bayesian optimization, iteratively selecting promising configurations based on past evaluations. A total of 50 trials were conducted, where each trial corresponds to a different hyperparameter configuration evaluated on the validation set. This was necessary to avoid biasing the search toward configurations optimal only for the default architecture (five layers), and instead allow the model complexity itself to be learned from the data.

Regularization strategies were incorporated to improve generalization and stabilize training. A learning rate scheduler was used to gradually decrease the step size during optimiza-

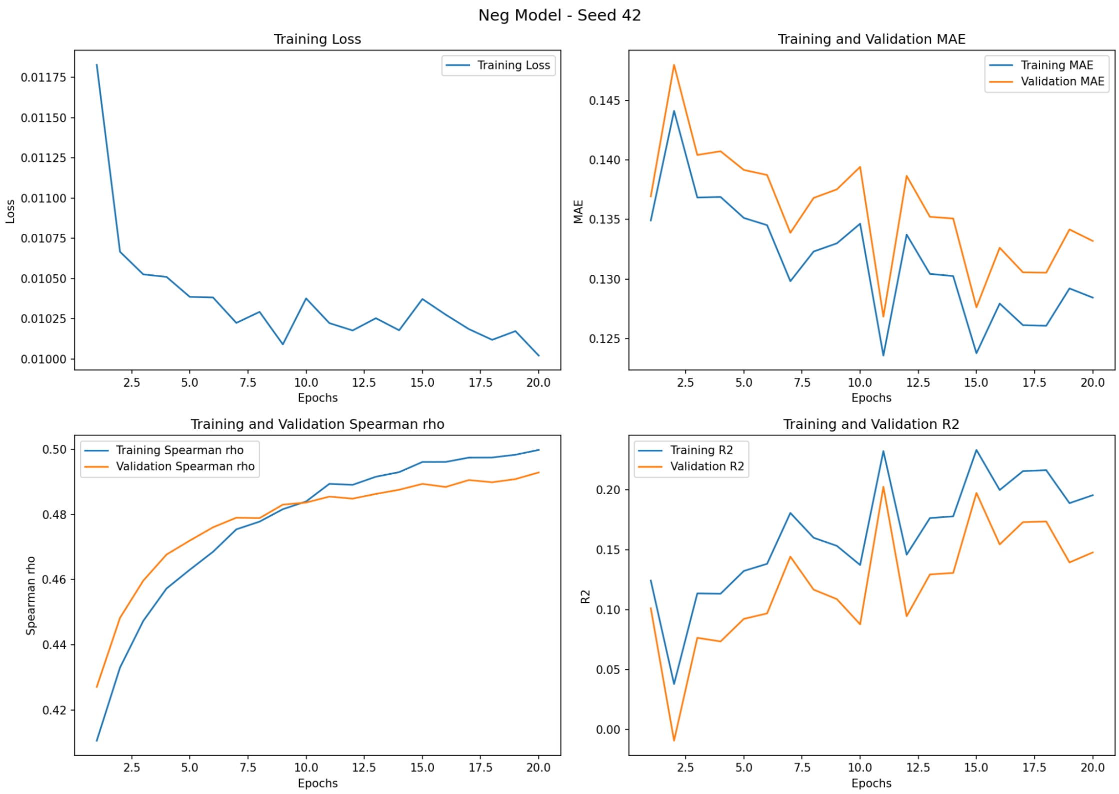


Figure 5.7: Training dynamics for the negative symptom model under default hyperparameters (seed 42).

tion, allowing faster convergence in early epochs and finer updates later. Early stopping was applied based on validation performance to prevent overfitting, halting training once improvements plateaued. Additionally, weight decay was introduced as an ℓ_2 penalty on model parameters to discourage overly complex representations and reduce variance. Together, these strategies aim to balance model flexibility with robustness in a relatively small and high-dimensional dataset.

The default and optimized hyperparameter configurations are provided in Appendix B.1, highlighting differences in model complexity and regularization between the two outcomes. The attention mechanism within each IGF block was excluded from tuning due to overfitting and fixed at its default configuration, while the global attention layers aggregating interaction- and feature-level representations were tuned.

Compared to the default configuration, the tuned models exhibit smoother and more stable training dynamics across all metrics as seen in Figure 5.8 and 5.9. Training and validation RMSE decrease consistently and remain closely aligned, indicating improved generalization and reduced overfitting.

Spearman correlation improves steadily with minimal divergence between training and validation curves, suggesting that the model captures robust rank-based relationships without overfitting to the training data. Notably, the tuned models achieve comparable or slightly improved correlation levels relative to the default setting, but with greater stability across epochs.

Similarly, validation R^2 shows more consistent improvement, with reduced fluctuations compared to the default model. While the overall magnitude of R^2 remains modest, the reduced variance across epochs indicates a more reliable fit.

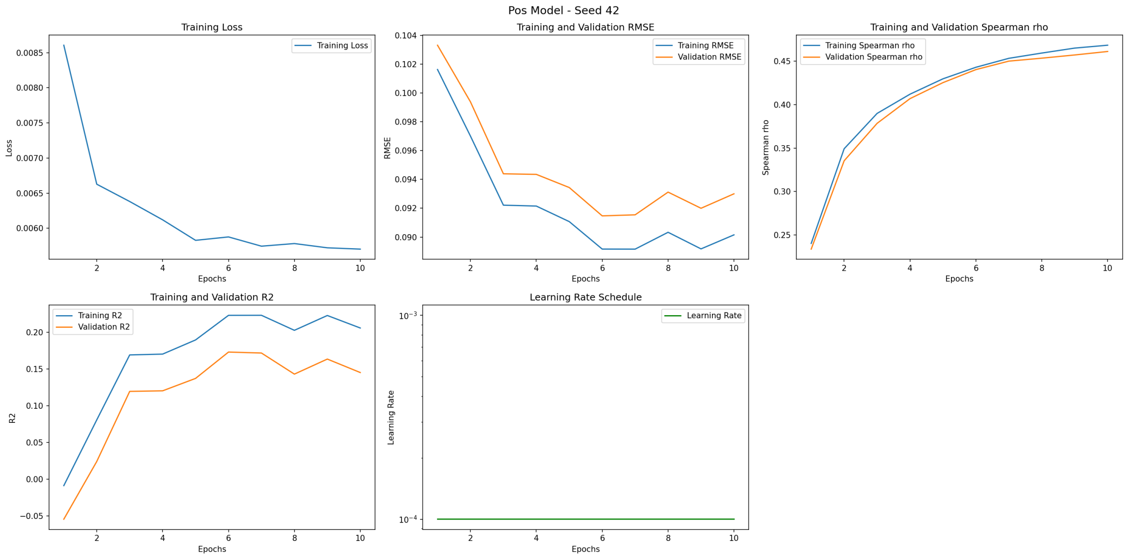


Figure 5.8: Training dynamics for the positive symptom model after hyperparameter tuning (seed 42).

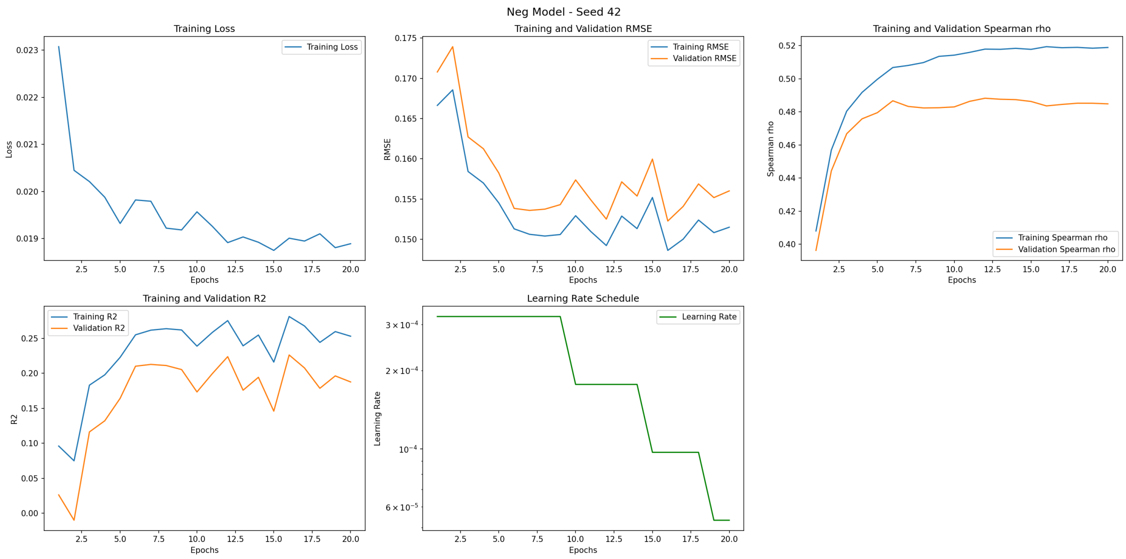


Figure 5.9: Training dynamics for the negative symptom model after hyperparameter tuning (seed 42).

The learning rate schedule further supports stable optimization, with stepwise decay enabling initial rapid learning followed by fine-tuning in later epochs.

Overall, hyperparameter tuning leads to more stable convergence and improved generalization, rather than large gains in absolute performance. This suggests that model performance is primarily limited by signal strength in the data rather than optimization inefficiencies. This pattern is consistent with high-dimensional observational data, where improved optimization primarily enhances stability rather than substantially increasing predictive power.

5.4 Model Comparison

Model performance was compared using calibration plots, which illustrate the relationship between predicted and observed values for both positive and negative outcomes (Figures 5.10, 5.11, and 5.12).

Across all models, a general pattern of regression toward the mean is observed, with predictions concentrated in a narrow range relative to the true values. This effect is particularly pronounced in the GLMM and GAMM models, where predictions remain close to the average outcome even at higher observed values.

The GLMM provides a reasonable baseline fit, with moderate alignment between predicted and observed values. The GAMM, despite allowing for non-linear effects, does not yield substantial improvements over the GLMM, suggesting that the added flexibility in modeling smooth effects of PRS does not significantly enhance predictive performance in this setting.

In contrast, the tuned DCMFNet model demonstrates improved performance across both outcomes, as reflected in higher R^2 and correlation metrics. More importantly, the calibration plots indicate that DCMFNet is better able to capture variability in the upper range of the outcome distribution. For the negative symptom model in particular, DCMFNet produces higher predicted values for individuals with elevated observed risk, whereas GLMM and GAMM tend to underestimate these cases.

This difference is critical, as the ability to distinguish higher-risk individuals is essential in this application. While all models perform comparably in the lower range of outcomes, DCMFNet exhibits greater sensitivity to tail behavior, suggesting that it is more effective in capturing complex, non-linear interactions that drive elevated symptom severity.

Overall, these results indicate that while statistical models provide interpretable and stable baselines, the deep learning approach offers improved flexibility and predictive capacity, particularly in identifying higher-risk individuals.

This aligns with the central research objective, indicating that explicitly modeling complex, non-linear interactions between genetic, environmental, and behavioral factors yields improved representation of schizophrenia risk, particularly in the upper tail of the outcome distribution.

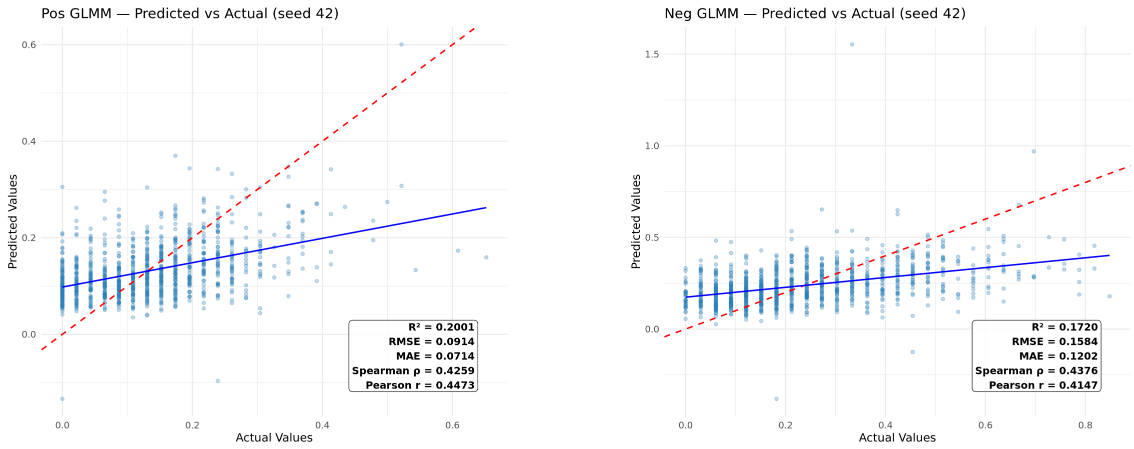


Figure 5.10: Calibration plots for GLMM models (positive and negative outcomes seed=42)

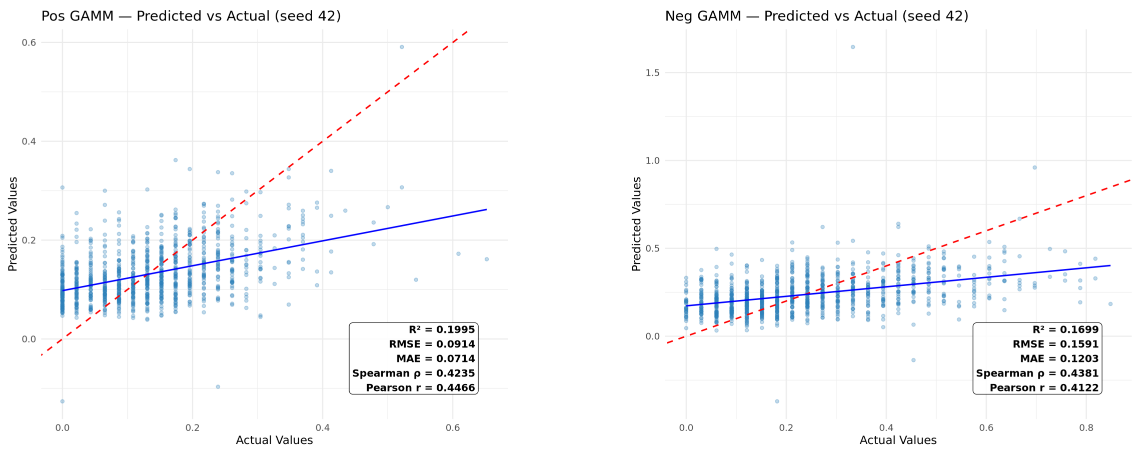


Figure 5.11: Calibration plots for GAMM models (positive and negative outcomes seed=42)

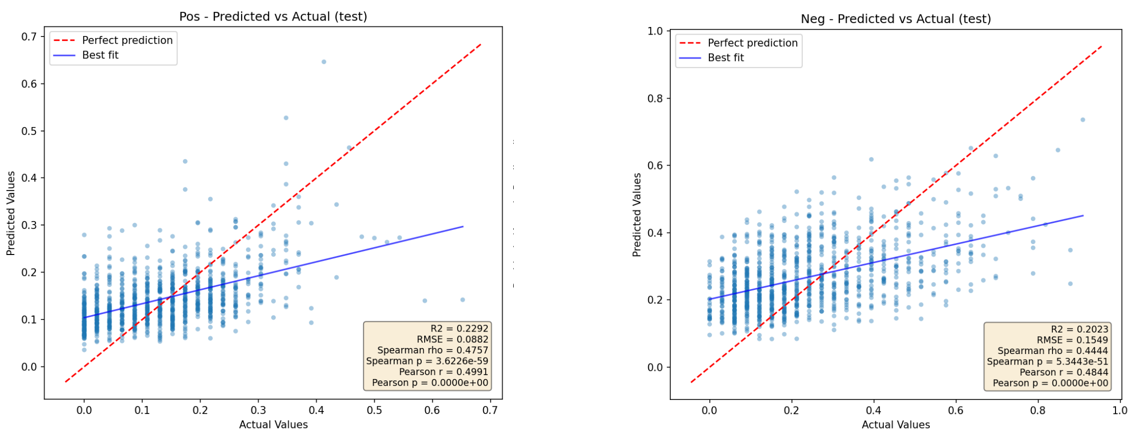


Figure 5.12: Calibration plots for DCMFNet models (positive and negative outcomes seed=42)

To assess robustness, model performance was averaged across multiple random seeds (42, 43, 44, 45, 46) as shown in Table 5.2. The results show consistent trends across runs, with DCMFNet outperforming both GLMM and GAMM in terms of explained variance, prediction error, and rank-based correlation.

Notably, GLMM and GAMM exhibit nearly identical performance across all metrics, indicating that the additional flexibility introduced through smooth terms does not materially improve predictive accuracy in this setting. In contrast, DCMFNet provides consistent gains, particularly in Spearman correlation, suggesting an improved ability to capture the ordering of risk. Improvements in R^2 remain modest, indicating that the primary advantage of DCMFNet lies in its ability to model complex relationships in the data rather than in substantially increasing explained variance.

The relatively low variance across seeds further indicates that these differences are stable and not driven by data partitioning.

Model	R^2	**RMSE**	**MAE**	**Spearman** ρ
GLMM (Pos)	0.181 ± 0.020	0.094 ± 0.002	0.073 ± 0.002	0.422 ± 0.021
GAMM (Pos)	0.180 ± 0.020	0.094 ± 0.002	0.073 ± 0.002	0.420 ± 0.021
DCMFNet (Pos)	$\mathbf{0.206 \pm 0.019}$	$\mathbf{0.090 \pm 0.001}$	$\mathbf{0.071 \pm 0.000}$	$\mathbf{0.450 \pm 0.019}$
GLMM (Neg)	0.179 ± 0.021	0.161 ± 0.005	0.124 ± 0.003	0.435 ± 0.013
GAMM (Neg)	0.178 ± 0.021	0.162 ± 0.005	0.124 ± 0.003	0.434 ± 0.010
DCMFNet (Neg)	$\mathbf{0.198 \pm 0.021}$	$\mathbf{0.156 \pm 0.002}$	$\mathbf{0.124 \pm 0.002}$	$\mathbf{0.451 \pm 0.013}$

Table 5.2: Model performance averaged across multiple random seeds (mean $\pm$ standard deviation).

5.5 ML – Benchmark Analysis

To provide additional context, we compared model performance with standard machine learning approaches, averaged across five random seeds.

Compared to standard machine learning models, performance differences are more nuanced. Several ML models (e.g., regularized linear models and kernel methods) achieve comparable or slightly higher performance in certain metrics. However, these gains are generally small and often fall within overlapping uncertainty ranges.

A clearer distinction emerges between the two outcomes. For both symptoms in Figure 5.13 and 5.14, model performance is tightly clustered across all methods suggesting that the limited power is with the data and not with the models.

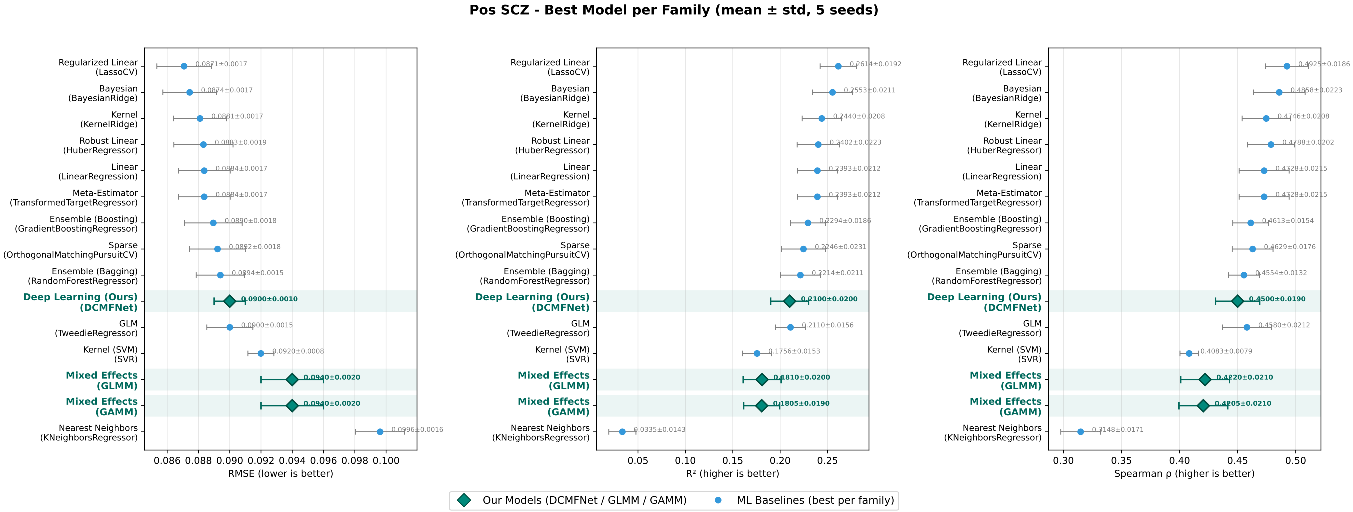


Figure 5.13: Model performance comparison by algorithm family for positive outcomes.

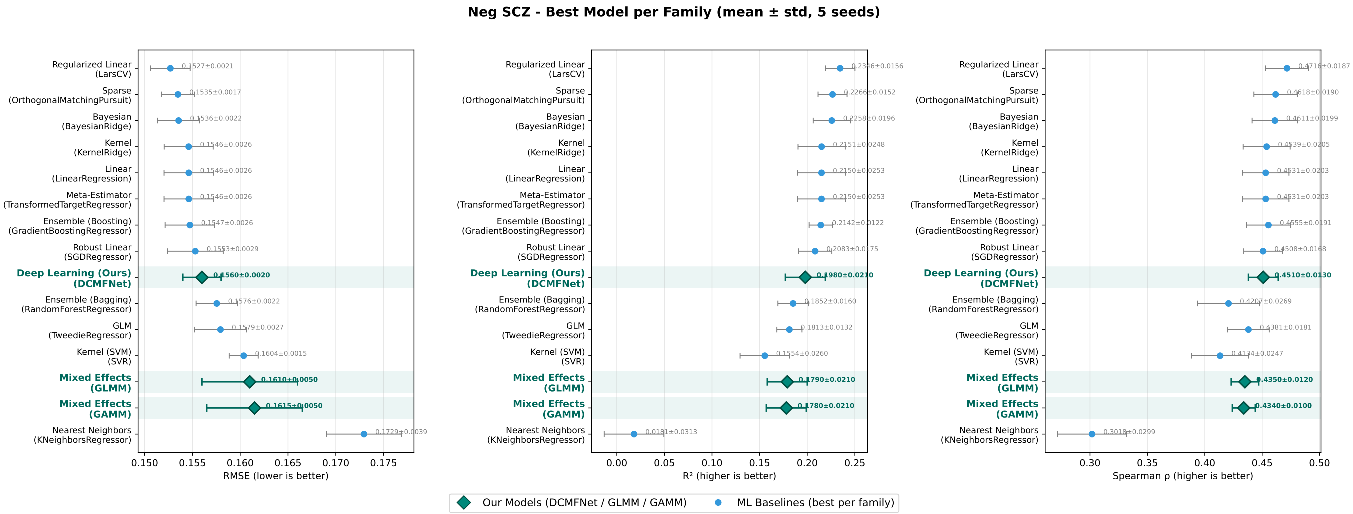


Figure 5.14: Model performance comparison by algorithm family for negative outcomes.

Instance-based methods such as K-nearest neighbors perform poorly and exhibit high variability, particularly for the negative outcome, suggesting that local similarity-based approaches are not well suited to the structured, high-dimensional nature of the data. In contrast, several standard machine learning models achieve stronger predictive performance.

However, these results should be interpreted with caution. For example, Bayesian models show strong performance but rely on distributional assumptions that may not hold in this setting. Similarly, linear regression performs relatively well, but this may reflect a tendency to predict mean values, as observed in the calibration plots for GLMM and GAMM in Figure 5.10 and Figure 5.11. Multiple testing would be required to account for potential false positives and better assess relative performance.

Overall, these results highlight a key trade-off. While several established machine learning models achieve stronger predictive performance, DCMFNet provides a balance between accuracy, stability, and interpretability. Its consistent improvement over GLMM and GAMM suggests that explicitly modeling interaction structure offers advantages over predefined or smoothly constrained approaches. These findings represent an initial step toward incorporating such architectures in this domain, rather than a replacement for established methods.

5.6 DCMFNet Model Analysis

5.6.1 Complexity Analysis

To better understand how interactions are represented within the DCMFNet architecture, we performed a complexity analysis by varying the depth of individual IGF blocks from 1 to 5 while holding others fixed at their default values. This optimal depth sweep was evaluated by minimizing RMSE, with the aim of identifying a model that provides a better fit. These results suggest that features requiring greater depth than the default reflect more complex interactions with substance use, whereas fewer layers indicate more direct relationships.

The results (Figures 5.15 and 5.16) reveal clear heterogeneity in the required interaction complexity across modalities averaged over 5 seeds.

Positive Model

For the positive symptom model, ADHD9 appears to exhibit a more complex interaction with substance use in predicting positive symptoms of schizophrenia, whereas all other features show more direct relationships.

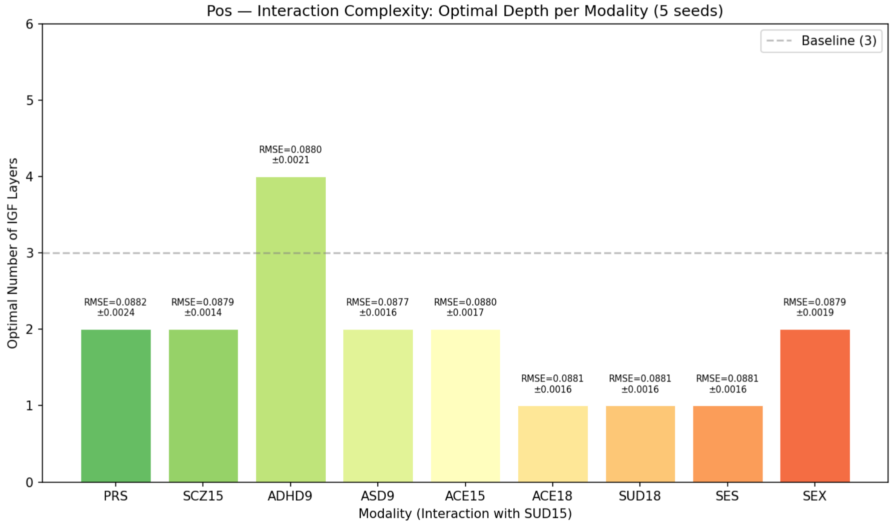


Figure 5.15: Optimal depth search for each modal block in the positive DCMFNet model averaged over 5 seeds.

Negative Model

For the negative symptom model, however, the pattern differs. PRS, SCZ15, and sex appear to exhibit more complex interactions with substance use, whereas ADHD9, ASD9, ACE18, SUD18, and SES show more direct relationships.

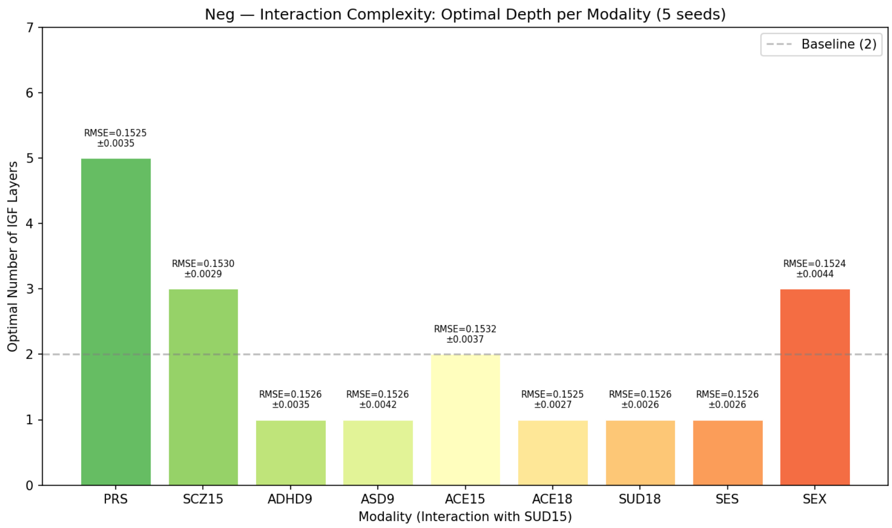


Figure 5.16: Optimal depth search for each modal block in the negative DCMFNet model averaged over 5 seeds.

An alternative approach jointly optimized the number of layers in each IGF block during hyperparameter tuning. However, this consistently led to overfitting on the test set, suggesting instability when optimizing a high-dimensional hyperparameter space given the sample size. We therefore retained the tuned hyperparameters under a fixed-depth configuration and used the present analysis to assess modality-specific interaction complexity in a more controlled manner.

This approach provides a level of granularity not available in GLMM or GAMM, revealing that different modalities operate at different levels of interaction complexity. However, further interpretation is limited, as the available data may be insufficient for the model to reliably learn complex interaction structures.

A quantitative comparison between default and optimized configurations is provided in Table 5.3, suggesting that adaptive depth improves performance for positive symptoms but not for negative symptoms. Based on these results, the best-performing configuration was selected for subsequent analysis.

Model	**Configuration**	**RMSE**	R^2	**Spearman** ρ
Positive	Optimal (Per-modality)	$\mathbf{0.0898 \pm 0.0015}$	$\mathbf{0.2145 \pm 0.0208}$	$\mathbf{0.4530 \pm 0.0150}$
Positive	Default (L=3)	0.0905 ± 0.0010	0.2023 ± 0.0143	0.4443 ± 0.0146
Negative	Optimal (Per-modality)	0.1569 ± 0.0014	0.1915 ± 0.0268	0.4348 ± 0.0256
Negative	Default (L=2)	$\mathbf{0.1569 \pm 0.0028}$	$\mathbf{0.1913 \pm 0.0319}$	$\mathbf{0.4449 \pm 0.0118}$

Table 5.3: Comparison between optimal per-modality and uniform depth configurations for DCMFNet (mean ± standard deviation across seeds).

5.6.2 Importance Analysis

To interpret the DCMFNet model, feature importance was assessed using attention weights from both independent and fused representations. Mean importance values were computed from the attention gates and averaged across five seeds.

Positive Model

Figure 5.17 shows that interactions involving ACE18, SES, SUD18, baseline symptoms at age 15, ASD9, and PRS exhibit relatively higher importance compared to other features, whereas sex shows comparatively lower importance. This finding is also consistent with the GLMM and GAMM results, which indicated a strong contribution of ACE18 in interaction with substance use shown in Figure 5.4.

Figure 5.18 indicates that ACE18, genetic confounding (batch×PC), PRS, and ADHD9 are relatively more important when considered independently, while sex, SES, and SUD15 show lower importance in the independent setting. Notably, the observation that SUD15 is more important in the interaction setting but less important independently.

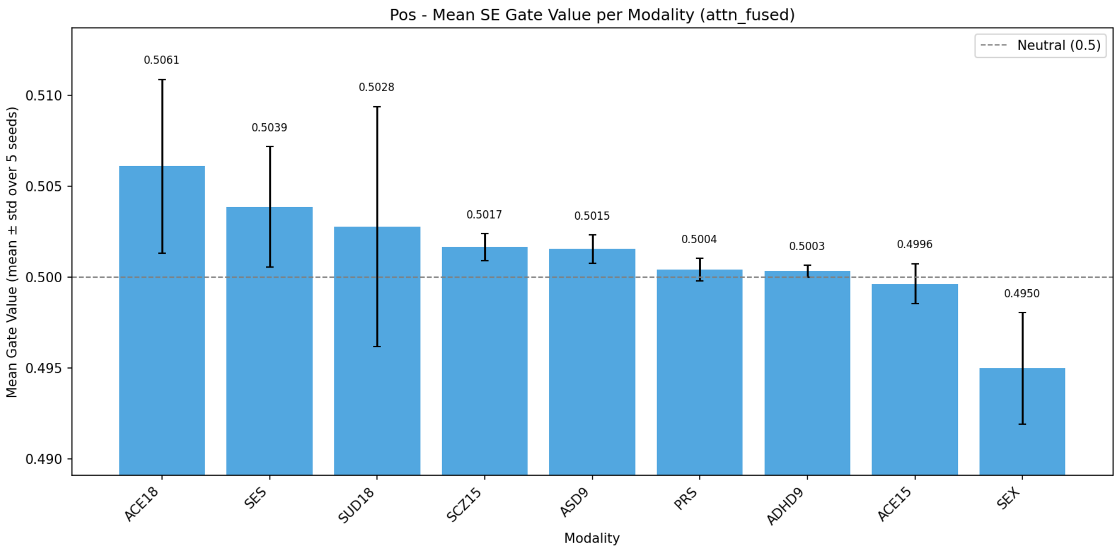


Figure 5.17: Mean Importance weights for each interaction feature in the positive DCMFNet model averaged over 5 seeds.

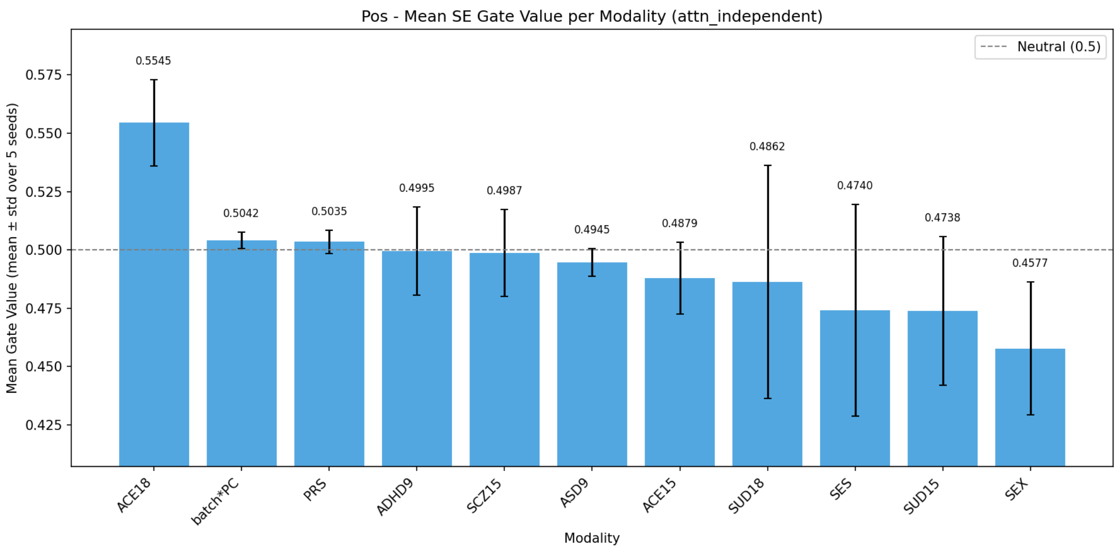


Figure 5.18: Mean Importance weights for each independent feature in the positive DCMFNet model averaged over 5 seeds.

Finally, Figure 5.19 presents the importance at the final layer, combining both independent and interaction streams. Here, the interaction stream appears more influential than the independent stream, supported by non-overlapping standard deviations. This finding aligns with the earlier observation from the generalized mixed models that positive symptoms are more strongly influenced by interaction effects shown in Figure 5.4.

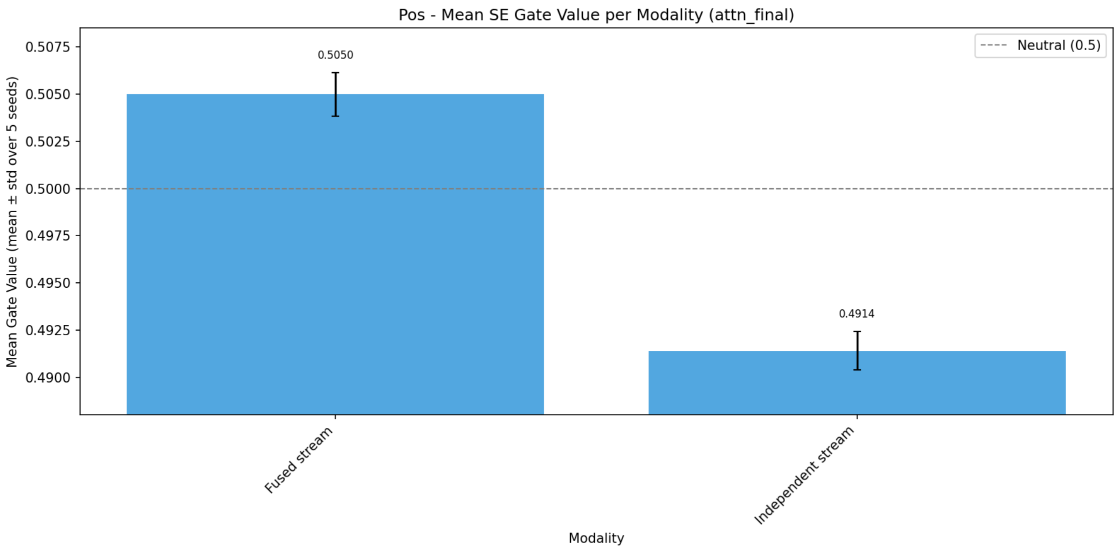


Figure 5.19: Mean Importance weights for final independent and interaction streams in the positive DCMFNet model averaged over 5 seeds.

Negative Model

For the interaction feature importance in the negative model in Figure 5.20, ACE18 appears to be the only feature with comparatively higher importance, while the remaining interactions show a diffuse pattern with values around 0.5. Notably, SUD18 exhibits relatively lower importance compared to the others.

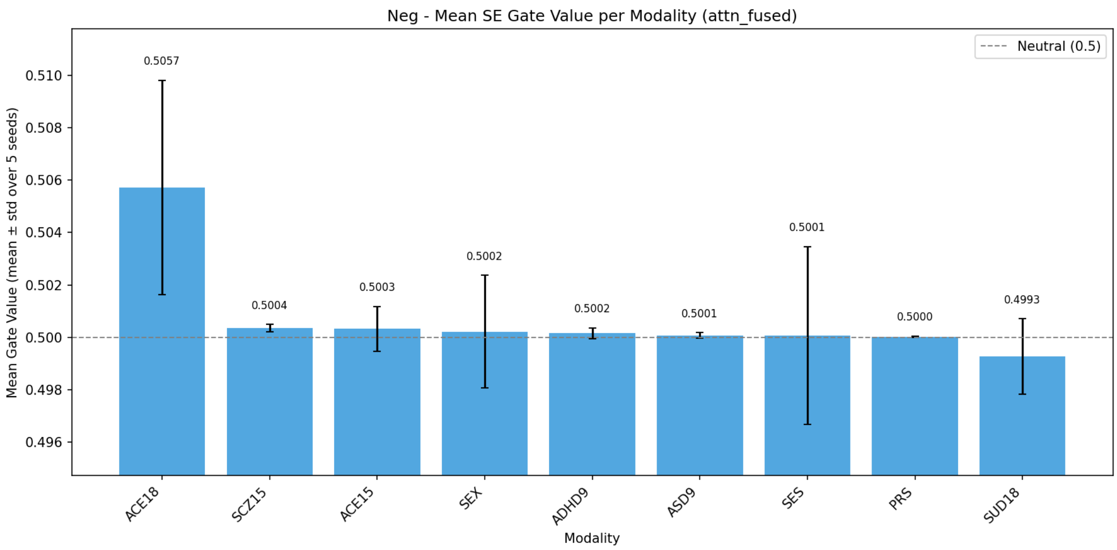


Figure 5.20: Mean Importance weights for each interaction feature in the negative DCMFNet model averaged over 5 seeds.

For independent features in Figure 5.21, a clearer pattern emerges, with ACE18, sex, baseline symptoms, and genetic confounding showing higher importance. This is consistent with the results from the generalized mixed models presented in Figure 5.5. In the negative model as well, SUD15 shows low importance as an independent feature, further suggesting that substance use does not act in isolation.

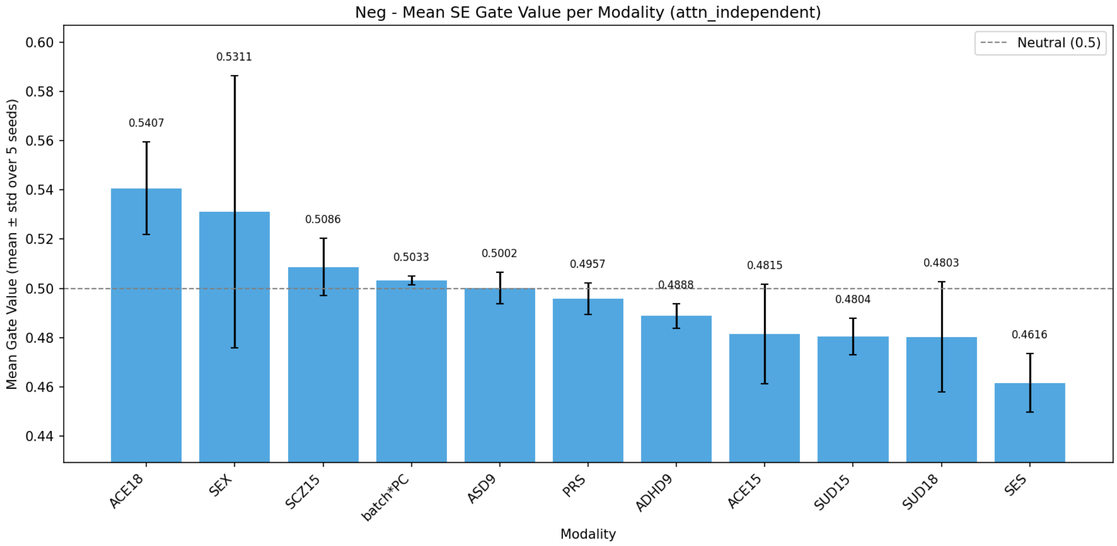


Figure 5.21: Mean Importance weights for each independent feature in the negative DCMFNet model averaged over 5 seeds.

Finally, Figure 5.22 presents the importance at the final layer, combining both independent and interaction streams for the negative symptoms. While the interaction stream appears more influential, the large and overlapping standard deviations limit clear differentiation between the two. This suggests that negative symptoms are less strongly influenced by interaction effects and instead exhibit a more diffuse structure compared to positive symptoms.

Overall, this importance analysis provides complementary insight into how DCMFNet learns feature contributions and interaction structure, highlighting both consistencies with and distinctions from generalized mixed models.

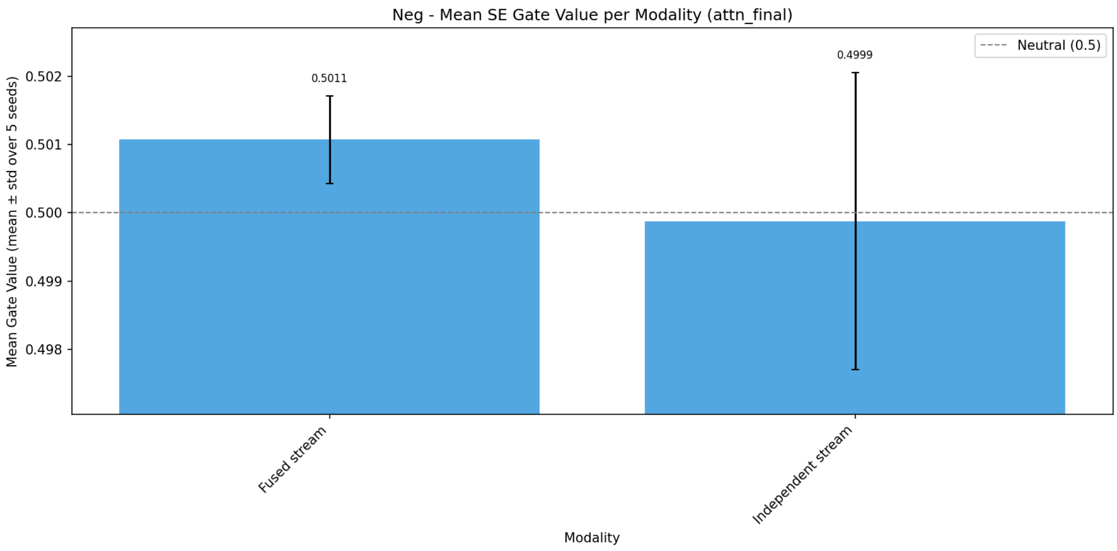


Figure 5.22: Mean Importance weights for final independent and interaction streams in the negative DCMFNet model averaged over 5 seeds.

Chapter 6

Discussion

6.1 Main Findings

This study examined how adolescent substance use interacts with genetic, environmental, and behavioral factors to influence psychotic symptoms, using both structured statistical models (GLMM, GAMM) and a deep learning approach (DCMFNet). Two main findings emerge, alongside a methodological observation.

First, psychotic risk—particularly for positive symptoms—appears to be organized through interactions rather than isolated effects, but this organization differs markedly across symptom domains. Across both GLMM and GAMM, the strongest associations for positive symptoms consistently arise from combinations of substance use with adverse childhood experiences (ACE) and genetic liability (PRS). These effects are selective rather than uniform, suggesting that substance use acts as a context-dependent amplifier whose influence is shaped by co-occurring vulnerabilities. Negative symptoms show a different structure: the dominant contributions arise from main effects—most notably PRS and sex—indicating a more distributed and less interaction-driven pattern of risk.

Second, the model's interpretability mechanisms supported this asymmetry. For positive symptoms, the interaction (fused) stream appeared more influential than the independent stream, with non-overlapping standard deviations across seeds. For negative symptoms, the two streams were largely indistinguishable, indicating that negative-symptom risk is less strongly shaped by interaction effects and exhibits a more diffuse structure than its positive counterpart. Together, these results indicate a genuine but modest difference in how the two symptom domains are organized.

Finally, model choice influenced not only predictive performance but also how well the underlying risk structure could be represented. Predictive power was modest across all model families—including strong machine-learning baselines—reflecting a genuine ceiling given the available features rather than a limitation of any single method. Within this ceiling, DCMFNet provided consistent, if modest, improvements over GLMM and GAMM, particularly in rank-based metrics, while also offering interpretable insight into which modalities and interactions drove prediction.

Taken together, these findings suggest that psychotic symptom risk is multifactorial and shaped by context-dependent interactions that differ across symptom domains, with positive symptoms more strongly interaction-driven than negative symptoms. These results should be read as associations within a cross-sectional, complete-case sample rather than causal or population-level effects; nonetheless, they indicate that greater model flexibility improves the representation of interaction structure, even where overall predictive gains remain limited.

6.2 Connection to Existing Literature

These findings support the long-standing view, established through twin studies by Sullivan et al. [3], of schizophrenia as a complex, multi-factorial disorder rather than one driven by any single dominant cause. Our results extend this picture by refining how these effects operate. In particular, cell-type specific PRS emerge as important contributors, consistent with Yao et al. [5], who connect genomic risk for psychiatric disorders to distinct brain cell types and regions. This highlights that genetic risk is not uniform but operates through biologically distinct pathways. Interactions between substance use and these cell-type PRS suggest that genetic vulnerability may be differentially amplified depending on the underlying cellular mechanisms.

Consistent with Robinson and Bergen [6], who describe environmental and genetic risk for psychosis as interrelated rather than independent, substance use in our models did not act in isolation but amplified risk in the presence of adverse childhood experiences and genetic liability. This aligns with evidence from Hall and Degenhardt [7] identifying substance use as a modifiable exposure linked to psychotic-disorder risk. However, in our analysis this amplification was selective rather than uniform, varying across substances and contexts—a more granular pattern than a single main effect would imply. ADHD- and ASD-related traits also emerged as important contributors, aligning with literature linking early neuro-developmental features to later psychosis risk, and reinforcing their role within broader interaction structures rather than as isolated predictors.

6.3 Implications for Mental Health Research

The implications of these findings are primarily clinical rather than methodological.

First, risk estimates should be interpreted as conditional rather than absolute. For an individual, factors such as substance use do not carry uniform risk, but depend on their co-occurrence with adverse experiences, baseline symptoms, and genetic vulnerability. This suggests that clinical assessment should move away from single-factor screening toward evaluating combinations of risk factors within a broader context.

Second, the differing structure of positive and negative symptoms implies that they may require distinct clinical strategies. Positive symptoms appear more sensitive to environmental and behavioral interactions, suggesting potential value in early behavioral or exposure-based interventions. In contrast, negative symptoms, being more stable and less interaction-driven, may require approaches focused on long-term support rather than short-term risk modification.

Overall, these results suggest that predictive models in mental health should prioritize identifying high-risk individuals and capturing context-dependent risk, rather than optimizing average prediction accuracy alone.

6.4 Strengths and Limitations

Each modeling approach contributes complementary strengths. GLMM and GAMM provide a clear and interpretable framework for estimating structured effects and accounting for dependencies such as the twin design. DCMFNet, in contrast, offers greater flexibility in modeling interaction structure, leading to improved predictive performance and stability. Importantly, the different modeling approaches do not provide competing explanations, but rather complementary perspectives on the same underlying system, each capturing different aspects of the relationship between predictors and outcome.

The main limitation of this study is data-related rather than methodological, specifically the effective sample size relative to model complexity. Although the dataset is large in absolute terms, high dimensionality, missingness, and stratification reduce the information available for learning complex interaction structures. This is reflected in the modest R^2 values across all models, indicating that a substantial proportion of variance remains unexplained.

A central tension in this work is between model flexibility and interpretability. GLMM and GAMM make all assumptions explicit and all effects directly readable, but cannot discover interaction structures beyond what is pre-specified. DCMFNet can discover complex patterns but requires post-hoc interpretation through attention weights, which are not directly comparable to regression coefficients. This suggests that in psychiatric research — where sample sizes are moderate, outcomes are noisy, and the interaction structure is unknown a priori — the most informative approach may not be selecting the best-performing model, but using multiple frameworks to triangulate the structure of risk.

6.5 Ethics

This study was conducted in accordance with the TRIPOD+AI guidelines [22], ensuring transparent reporting and evaluation of predictive models. The use of observational data requires careful interpretation, particularly when modeling sensitive outcomes such as mental health risk.

All analyses were performed on de-identified data, and no attempt was made to derive individual-level predictions for clinical use. The models are intended to support understanding of population-level patterns rather than decision-making at the individual level.

Special care was taken to avoid deterministic interpretations of genetic risk. PRS variables reflect probabilistic associations and should not be interpreted as fixed or predictive at the individual level. Similarly, associations with behavioral or environmental exposures do not imply causation or inevitability.

6.6 Future Work

Several directions follow naturally from this work. The internal representations learned by DCMFNet — particularly the per-layer fusion outputs — can be further analyzed to understand how interaction signals build across layers, potentially revealing which stages of fusion contribute most to risk differentiation. Random effects from the mixed models can be explored to identify meaningful twin-pair subgroups or outliers, which may point to unmeasured confounders or distinct risk profiles. Multiple imputation should be investigated as an alternative to complete case analysis, given that the current approach may introduce selection bias if missingness is not completely at random.

Validation in an independent cohort — ideally from a different population — would establish whether the identified interaction structures generalize beyond Swedish twins. High-performing machine learning models identified through the benchmark analysis can be systematically tuned to provide stronger baselines. Multiple testing correction can be applied across all models, including benchmark machine learning models, to control the Type I error rate.

Finally, extending DCMFNet to explicitly incorporate the twin-pair structure — analogous to the random intercept in GLMM and GAMM — would address a current limitation where the deep learning model does not account for within-pair dependence. More broadly, this work demonstrates that it is now feasible to model structured interactions between substance use and multiple domains within a unified framework. Future studies can extend this approach to capture richer interaction patterns across features, enabling a more detailed understanding of how complex risk factors combine in schizophrenia and potentially other multi-factorial psychiatric disorders.

Code is available at: `https://github.com/sil-pixel/Thesis`

Chapter 7

Conclusions

This work set out to model schizophrenia risk through interactions, and the results refine how that problem should be framed. Risk in this cohort is not well explained by individual factors alone, nor uniformly by interactions. Instead, a small set of main effects—adverse childhood experiences, prior symptoms, and polygenic liability—forms a stable baseline, while substance use modifies this risk selectively in combination with adversity and genetic factors.

The clearest distinction emerges between symptom domains. Positive symptoms show concentrated interaction effects, particularly involving substance use with ACE and cell-type PRS, with the DCMFNet interaction stream contributing more strongly than the independent stream. In contrast, negative symptoms are largely driven by main effects, with limited differentiation between interaction and independent contributions. Overall, substance use acts as a context-dependent amplifier for positive symptoms, but plays a more limited role for negative symptoms.

No single model was uniformly best, and each contributed something distinct. GLMM and GAMM made the structure interpretable, showing which factors and interactions carried weight. DCMFNet represented those relationships more flexibly, yielding consistent if modest gains in rank-based metrics and clearer separation for higher-risk individuals. The contribution is therefore methodological as much as substantive: combining interpretable baselines with a flexible interaction model gave a fuller view of risk structure than either alone, even though predictive power was modest across all approaches—including strong machine-learning baselines—reflecting a genuine ceiling in the available features.

These results should be read as associations within a cross-sectional, complete-case twin sample, not as causal or population-level effects. Within those limits, they suggest that progress in modeling schizophrenia risk may depend less on discovering new predictors than on better representing how known factors—genetic, environmental, and behavioral—combine across adolescent development, and on recognizing that this combination differs between positive and negative symptom domains.

Chapter 8

Declaration of AI

Artificial intelligence tools were used to assist in code development and refinement of written text. All outputs were critically reviewed and adapted by the author. The ideas, interpretations, and conclusions presented in this work are the author's own.

References

[1] A. J. Robison, Katharine N. Thakkar, and Vaibhav A. Diwadkar. "Cognition and Reward Circuits in Schizophrenia: Synergistic, Not Separate". In: *Biological Psychiatry* 87.3 (2020), pp. 204–214. DOI: 10.1016/j.biopsych.2019.09.021. URL: https://doi.org/10.1016/j.biopsych.2019.09.021.

[2] Na Zhan et al. "The genetic basis of onset age in schizophrenia: evidence and models". In: *Frontiers in Genetics* Volume 14 - 2023 (2023). ISSN: 1664-8021. DOI: 10.3389/fgene.2023.1163361. URL: https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2023.1163361.

[3] Neale MC Sullivan PF Kendler KS. "Schizophrenia as a Complex Trait: Evidence From a Meta-analysis of Twin Studies". In: *Arch Gen Psychiatry* (2003). DOI: doi:10.1001/archpsyc.60.12.1187.

[4] Enkhmaa Luvsannyam et al. "Neurobiology of Schizophrenia: A Comprehensive Review". In: *Cureus* 14.4 (2022), e23959. DOI: 10.7759/cureus.23959. URL: http://dx.doi.org/10.7759/cureus.23959.

[5] Shuyang Yao et al. "Connecting genomic results for psychiatric disorders to human brain cell types and regions reveals convergence with functional connectivity". In: *Nature Communications* 16.1 (2025), p. 395. DOI: 10.1038/s41467-024-55611-1. URL: https://doi.org/10.1038/s41467-024-55611-1.

[6] Natassia Robinson and Sarah E. Bergen. "Environmental Risk Factors for Schizophrenia and Bipolar Disorder and Their Relationship to Genetic Risk: Current Knowledge and Future Directions". In: *Frontiers in Genetics* Volume 12 - 2021 (2021). ISSN: 1664-8021. DOI: 10.3389/fgene.2021.686666. URL: https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2021.686666.

[7] Wayne Hall and Louisa Degenhardt. "Cannabis use and the risk of developing a psychotic disorder." eng. In: *World Psychiatry* 7.2 (2008), pp. 68–71. ISSN: 1723-8617 (Print); 1723-8617 (Linking). DOI: 10.1002/j.2051-5545.2008.tb00158.x.

[8] Camila Maria Severi Martins et al. "Emotional Abuse in Childhood Is a Differential Factor for the Development of Depression in Adults". In: *The Journal of Nervous and Mental Disease* 202.11 (2014).

[9] Ciaran Shannon et al. "M31. THE ASSOCIATION BETWEEN BULLYING VICTIMIZATION AND PSYCHOSIS AND THE MEDIATING ROLE OF PEER AND

FAMILY SOCIAL SUPPORT". In: *Schizophrenia Bulletin* 46.Supplement$_1$ (May 2020), S146–S146.

[10] Amandeep Jutla, Jennifer Foss-Feig, and Jeremy Veenstra-VanderWeele. "Autism spectrum disorder and schizophrenia: An updated conceptual review". In: *Autism Research* 15.3 (2022), pp. 384–412. DOI: https://doi.org/10.1002/aur.2659. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/aur.2659. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/aur.2659.

[11] S. Dalsgaard et al. "Association between Attention-Deficit Hyperactivity Disorder in childhood and schizophrenia later in adulthood". In: *European Psychiatry* 29.4 (2014), pp. 259–263. DOI: 10.1016/j.eurpsy.2013.06.004.

[12] Esben Agerbo et al. "Polygenic Risk Score, Parental Socioeconomic Status, Family History of Psychiatric Disorders, and the Risk for Schizophrenia: A Danish Population-Based Study and Meta-analysis". In: *JAMA Psychiatry* 72.7 (July 2015), pp. 635–641. ISSN: 2168-622X. DOI: 10.1001/jamapsychiatry.2015.0346. eprint: https://jamanetwork.com/journals/jamapsychiatry/articlepdf/2211892/yoi150019.pdf. URL: https://doi.org/10.1001/jamapsychiatry.2015.0346.

[13] Brooke Carter et al. "Sex and gender differences in symptoms of early psychosis: a systematic review and meta-analysis". In: *Archives of Women's Mental Health* 25.4 (2022), pp. 679–691.

[14] Kimberly Siletti et al. "Transcriptomic diversity of cell types across the adult human brain". In: *Science* 382.6667 (2023), eadd7046. DOI: 10.1126/science.add7046. eprint: https://www.science.org/doi/pdf/10.1126/science.add7046. URL: https://www.science.org/doi/abs/10.1126/science.add7046.

[15] Ashley E. Tate et al. "Predicting mental health problems in adolescence using machine learning techniques". In: *PLOS ONE* 15.4 (Apr. 2020), pp. 1–13. DOI: 10.1371/journal.pone.0230389. URL: https://doi.org/10.1371/journal.pone.0230389.

[16] Zhen Huang et al. "DCMFNet: Deep Cross-Modal Fusion Network for Different Modalities with Iterative Gated Fusion". In: *Association for Computing Machinery*. GI '24 (2024). DOI: 10.1145/3670947.3670956. URL: https://doi.org/10.1145/3670947.3670956.

[17] Tian Ge et al. "Polygenic prediction via Bayesian regression and continuous shrinkage priors". In: *Nature Communications* 10.1 (2019), p. 1776. DOI: 10.1038/s41467-019-09718-5. URL: https://doi.org/10.1038/s41467-019-09718-5.

[18] Han Wang. "The association between mechanism-informed polygenic risk score and schizophrenia and treatment resistance". Master's thesis. Stockholm, Sweden: Karolinska Institutet, 2025.

[19] Zhi-Hua Zhou. "Ensemble Methods: Foundations and Algorithms (1st ed.)" In: *Chapman and Hall/CRC* (2012). DOI: https://doi.org/10.1201/b12207.

[20] Jie Hu et al. "Squeeze-and-Excitation Networks". In: (2019). arXiv: 1709.01507 [cs.CV]. URL: https://arxiv.org/abs/1709.01507.

[21] Shankar Rao Pandala. *LazyPredict*. Version 0.3.0. Mar. 2026. URL: https://github.com/shankarpandala/lazypredict.

[22] Gary S Collins et al. "TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods". In: *BMJ* 385 (2024). DOI: 10.1136/bmj-2023-078378. eprint: https://www.bmj.com/content/385/bmj-2023-078378.full.pdf. URL: https://www.bmj.com/content/385/bmj-2023-078378.

Appendix A

SHAP Clustering

You can cluster your data with the help of Shapley values. The goal of clustering is to find groups of similar instances. SHAP clustering works by clustering the Shapley values of each instance. This means that you cluster instances by explanation similarity. All SHAP values have the same unit – the unit of the prediction space. To better understand the behavior of features from different perspectives, clustering was performed using two combinations of distance and linkage metrics.

1. First, Euclidean distance with Ward linkage was used to group features based on the magnitude of their SHAP contributions, allowing the identification of features with similar levels of importance while minimizing within-cluster variation.

2. Second, correlation distance with complete linkage was applied to group features based on similarity in their SHAP contribution patterns, regardless of scale, while ensuring that clusters remain tightly defined.

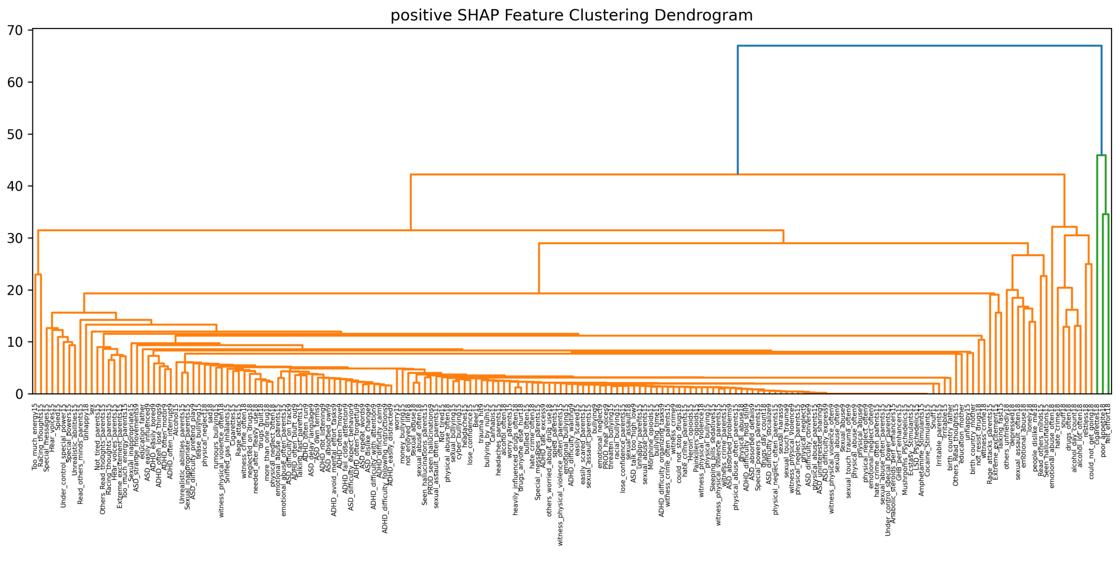


Figure A.1: SHAP Clustering of Positive LightGBM model using Ward linkage

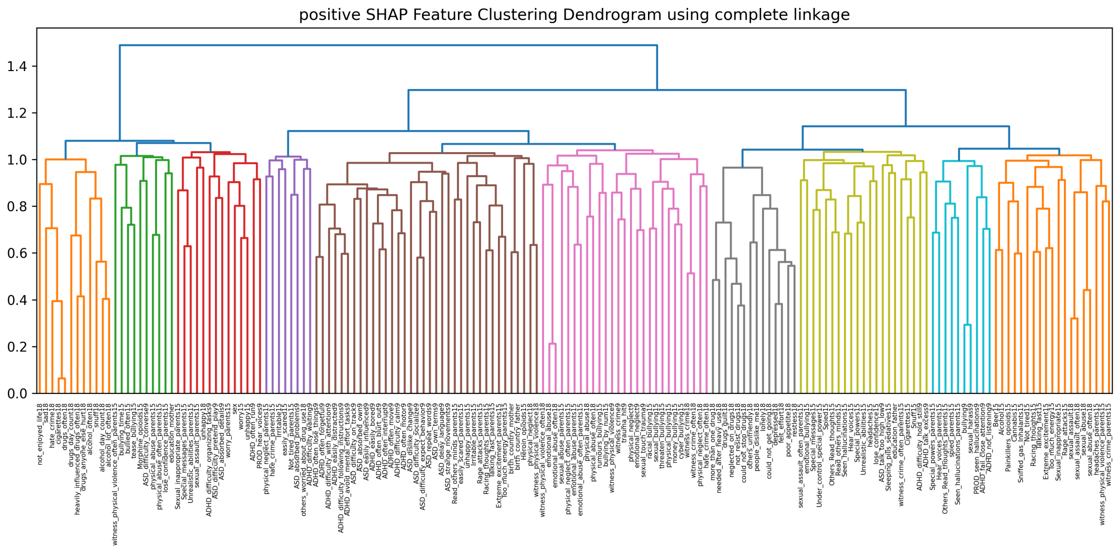


Figure A.2: SHAP Clustering of Positive LightGBM model using Complete linkage

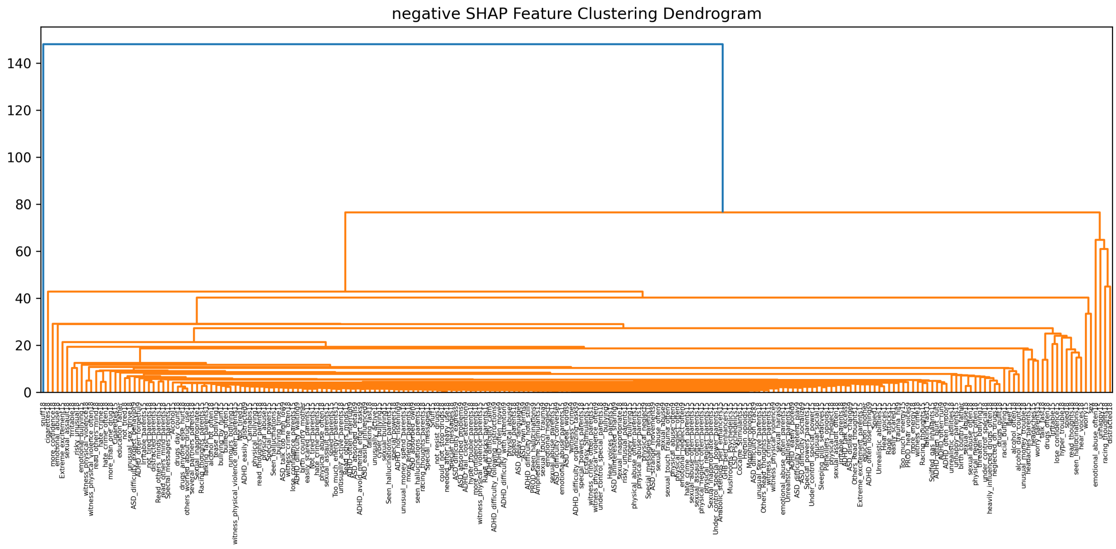


Figure A.3: SHAP Clustering of Negative LightGBM model using Ward linkage

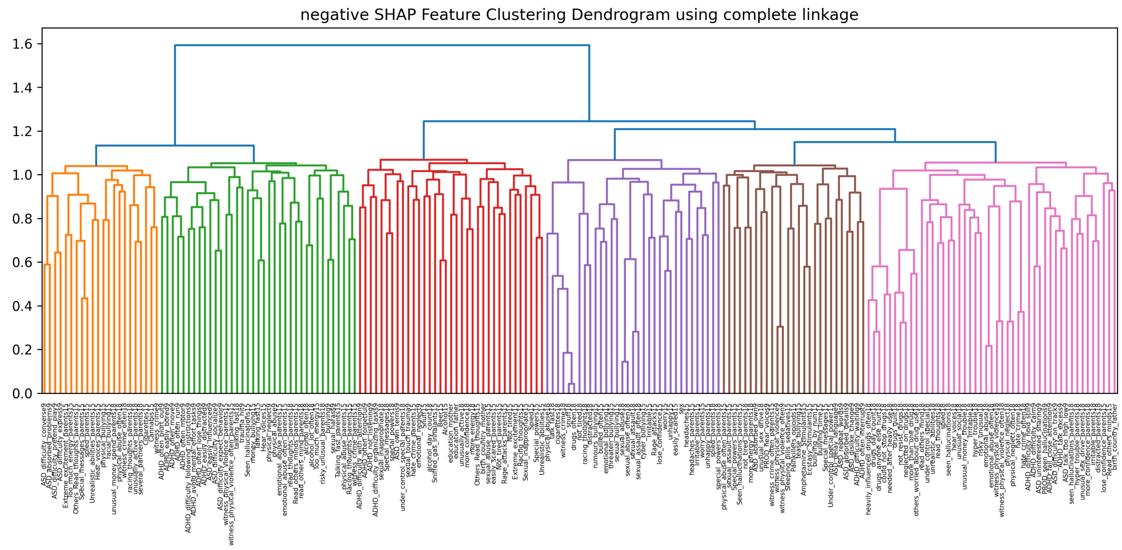


Figure A.4: SHAP Clustering of Negative LightGBM model using Complete linkage

Appendix B

DCMFNet Model Training

B.1 Hyperparameters

The tuned models exhibit lower weight decay, adjusted learning rates, and increased architectural flexibility, indicating a shift toward reduced regularization and greater capacity to capture complex interactions.

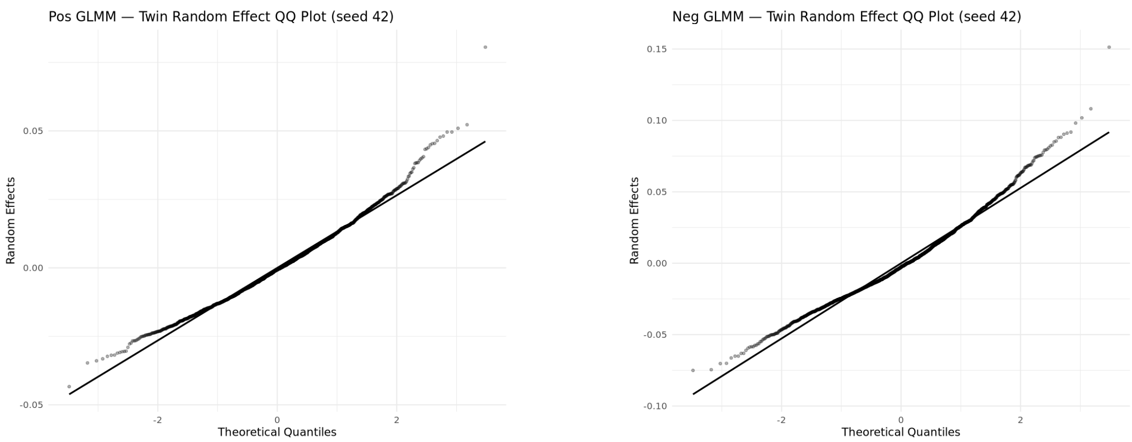
Hyperparameter	**Default**	**Positive Model**	**Negative Model**
Learning rate	1.82×10^{-4}	1.00×10^{-4}	3.22×10^{-4}
Batch size	8	4	6
Epochs	15	23	20
Optimiser	Adam	Adam	Adam
Weight decay	1.00×10^{-3}	2.95×10^{-5}	1.79×10^{-4}
Number of layers	5	3	2
Dropout	0.30	0.234	0.326
SE reduction	2	6	5
Min hidden dimension	8	13	5
Base loss	Huber	Huber	Huber
Huber δ	0.05	0.086	0.116
Focal γ	2.0	1.32	2.62
Number of bins	10	13	10
Scheduler patience	3	4	2
Scheduler factor	0.5	0.432	0.549
Early stopping patience	5	5	7
Seed	42	42	42

Table B.1: Comparison of default and tuned hyperparameters for DCMFNet models.

Appendix C

Results

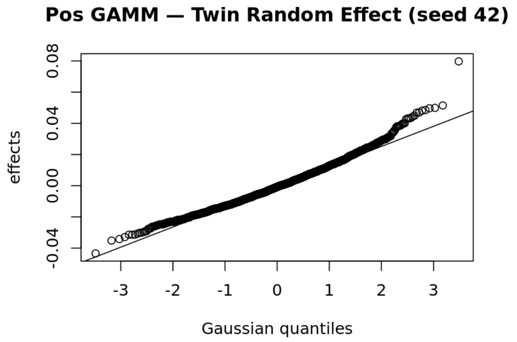
C.1 Random Effects



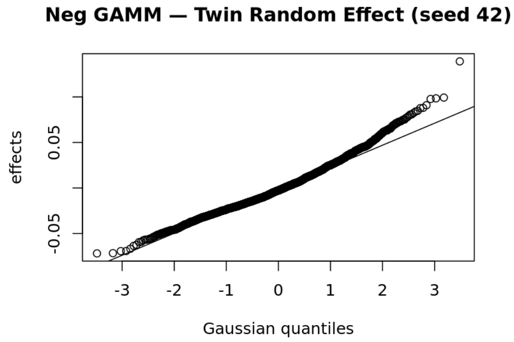
(a) Random effect of Twin in Positive GLMM Model at seed 42

(b) Random effect of Twin in Negative GLMM Model at seed 42

Figure C.1: Random effect of Twin in GLMM Models at seed 42



(a) Random effect of Twin in Positive GAMM Model at seed 42



(b) Random effect of Twin in Negative GAMM Model at seed 42

Figure C.2: Random effect of Twin in GAMM Models at seed 42

QQ plots of the twin-level random effects indicate approximate normality for both GLMM and GAMM models as seen in Figures C.1 and C.2. While most points align closely with the theoretical quantiles, mild deviations are observed in the tails, particularly for higher quantiles, suggesting the presence of a small number of twin pairs with larger-than-expected effects.

These deviations are limited in magnitude and do not indicate substantial model misspecification. The Gaussian random intercept therefore provides a reasonable approximation for capturing within-pair dependence in this dataset.

The smooth terms for cell-type specific PRS (Figures C.3– C.4) show largely weak and near-linear effects, with most confidence intervals overlapping zero across the observed range. This suggests limited non-linear contribution of PRS components when modeled independently. Some components (e.g., interneuron-related scores) exhibit mild directional trends, but overall effects remain small and uncertain.

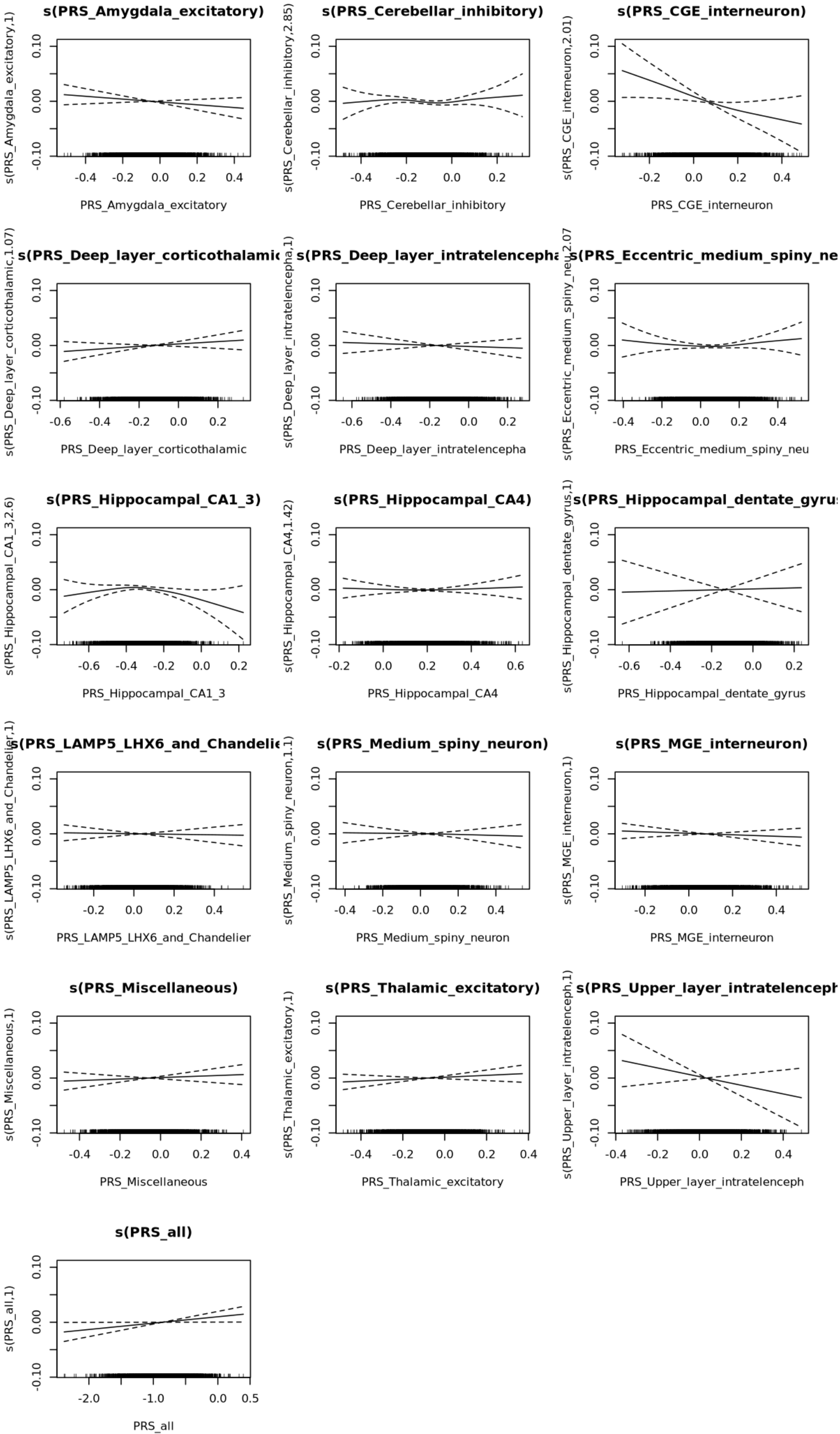


Figure C.3: Random effect of cell-type PRS smooth terms in Positive GAMM Model

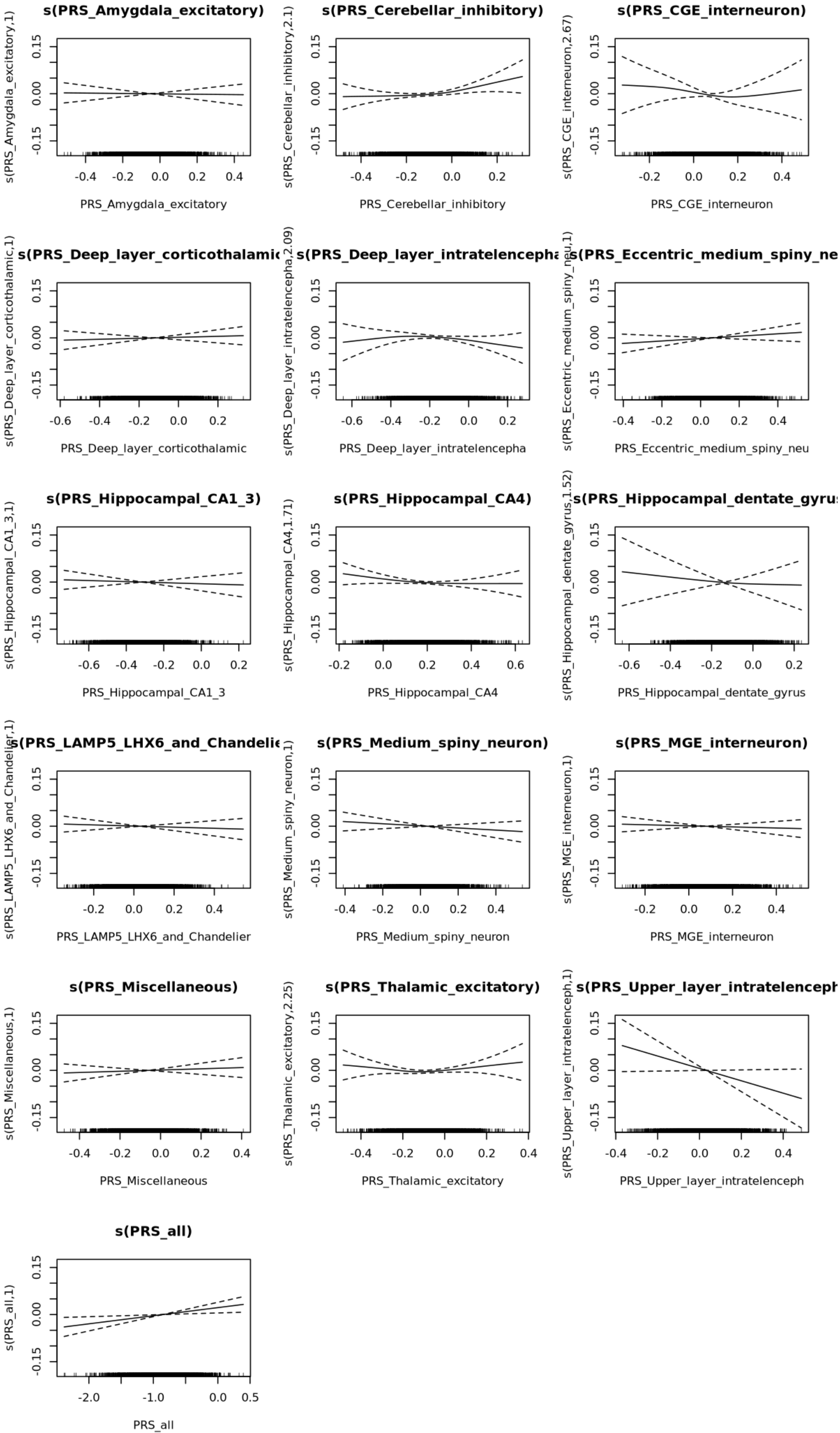


Figure C.4: Random effect of cell-type PRS smooth terms in Negative GAMM Model

C.2 GAMM Main Effects

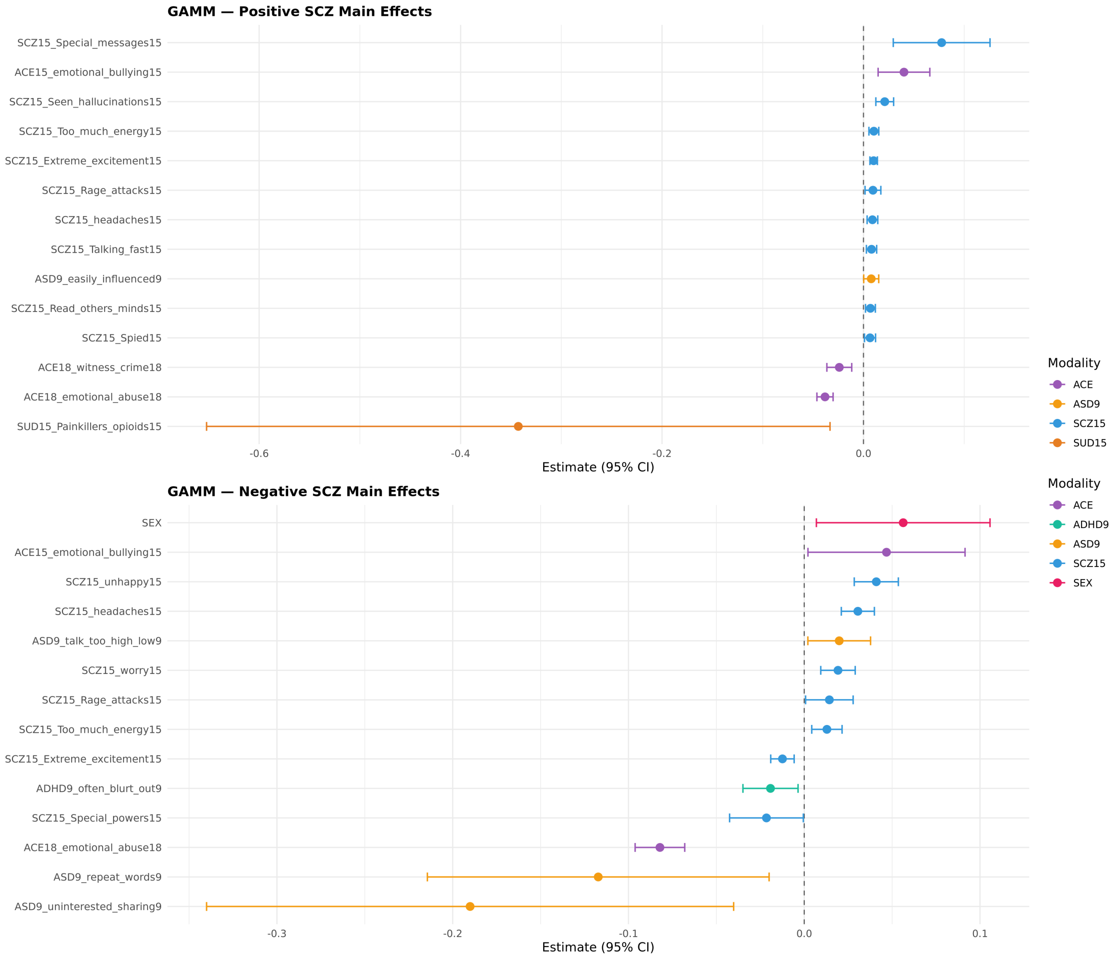


Figure C.5: Main effects in GAMM models for positive and negative outcomes at seed 42 using regression coefficients.